Analysis on Metric and Function Spaces

— ***EYNTKA*** **Series** —

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Introduction

This book lifts analysis off the real line. The single-variable theory developed in *Elementary Analysis* [1] is stated for functions on intervals, and almost every argument in it turns on two features of $\mathbb{R}$: that it carries a distance, and that it is complete with respect to that distance. Neither feature is special to the real line. Once they are isolated as axioms, the same arguments run on spaces of sequences, spaces of functions, and spaces of far stranger objects, and they run without change.

That abstraction is the subject here. The book develops metric spaces and completeness, the contraction principle and its fixed points, the several inequivalent-looking notions of compactness that coincide for metric spaces, and uniform convergence. It then turns to the space $C(K)$ of continuous functions on a compact space, where the payoff comes: Stone-Weierstrass, which says polynomials are enough; Arzela-Ascoli, which says equicontinuity and boundedness are enough to extract convergent subsequences; and the Baire category theorem, which manufactures objects nobody can write down.

These are the theorems the later analysis volumes use rather than reprove, and this book is the series' single home for them.

Part I

Metric Spaces

Single-variable analysis proves its theorems about $\mathbb{R}$, but it rarely uses more of $\mathbb{R}$ than two facts: that any two points have a distance between them, and that a sequence whose terms crowd together has somewhere to converge to. This Part promotes those two facts to axioms and asks how much of the theory survives.

The answer is: nearly all of the vocabulary, and rather less of the substance. Chapter 1 installs the vocabulary and finds that open sets, closure, convergence and continuity all transfer without a word of change, and that the three competing definitions of continuity stay equivalent. Chapter 2 isolates the second fact as completeness and shows it is a strictly metric notion, not a topological one. **??** extracts the sharpest single consequence of completeness, the contraction principle, which converts a shrinking map into an existence-and-uniqueness theorem. **??** confronts the one notion that does not transfer intact: in $\mathbb{R}^n$ compactness means closed and bounded, in a general metric space it does not, and sorting out what it does mean occupies the chapter.

Throughout, four spaces recur and are worth becoming familiar with: $\mathbb{R}^n$, the space $C[0,1]$ of continuous functions under the distance $\sup_x |f(x) - g(x)|$, the sequence space ℓ^2, and the Cantor set. Each new definition is worth testing against all four.

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Metric Spaces and Their Topology

The ϵ-δ definition of continuity mentions the real numbers only through the quantity $|x - y|$. So does the definition of a convergent sequence, of an open interval, of a limit point, and of a Cauchy sequence. Replacing $|x - y|$ by an abstract distance $d(x, y)$ therefore changes nothing in those definitions, and this chapter carries out the replacement and checks what it costs in the theorems.

The four sections install the vocabulary in order. Section 1.1 gives the axioms and a stock of examples, several of which are spaces whose points are functions or sequences rather than numbers. Section 1.2 builds open and closed sets, closure and convergence out of the distance alone. Section 1.3 proves that the ϵ-δ, sequential and preimage-of-open definitions of continuity remain equivalent, and separates continuity from its uniform and Lipschitz strengthenings. Section 1.4 handles the three constructions every later chapter uses without comment: passing to a subspace, forming a product, and replacing a metric by an equivalent one.

1.1 Metrics and Examples

Definition 1.1.1: Metric Space

A *metric* on a set X is a function $d\colon X \times X \to \mathbb{R}$ such that for all $x, y, z \in X$:

(i) $d(x, y) = 0$ if and only if $x = y$;

(ii) $d(x, y) = d(y, x)$;

(iii) $d(x, z) \le d(x, y) + d(y, z)$.

The pair (X, d) is a *metric space*, and the elements of X are its *points*. Where the metric is clear from context, X alone names the space.

Nothing in the definition requires that distances be positive, because they cannot help being so.

Lemma 1.1.2: Distances Are Nonnegative

If d is a metric on X then $d(x, y) \ge 0$ for all $x, y \in X$.

***Proof*:**
Apply the triangle inequality with $z = x$ and then symmetry:

$$0 = d(x, x) \le d(x, y) + d(y, x) = 2d(x, y)$$

so $d(x, y) \ge 0$, as we sought to show.

The first axiom is the one that is easy to weaken by accident, and the cost of weakening it is severe. Requiring only $d(x, x) = 0$, rather than the equivalence, admits the function $d \equiv 0$ on any set whatsoever: it is symmetric, it satisfies the triangle inequality, and every point is at distance zero from itself. Such a function is a *pseudometric*. Under it every sequence converges to every point at once, so limits are not unique (proposition 1.2.6), and the entire theory of section 1.2 collapses. The "only if" is what forces distinct points to be separated by some positive distance, and every result in this book uses it.

The examples below are the stock this book draws on. The first two are the familiar ones; the rest are spaces whose points are not numbers, and they are the reason the abstraction is worth making.

Example 1.1.3: The Standard Metrics on $\mathbb{R}^n$
On $\mathbb{R}^n$ define, for $x = (x_1, \dots, x_n)$ and $y = (y_1, \dots, y_n)$,

$$d_1(x, y) = \sum_{i=1}^{n} |x_i - y_i| \qquad d_2(x, y) = \left(\sum_{i=1}^{n} |x_i - y_i|^2 \right)^{1/2} \qquad d_\infty(x, y) = \max_{1 \le i \le n} |x_i - y_i|$$

the *taxicab*, *Euclidean* and *sup* metrics. For d_1 and d_∞ the axioms are immediate from the corresponding facts about $|\cdot|$ on $\mathbb{R}$: symmetry and the vanishing condition hold coordinatewise, and for the triangle inequality,

$$|x_i - z_i| \le |x_i - y_i| + |y_i - z_i|$$

summing gives the claim for d_1, while taking the maximum over i of the right-hand side, which dominates each summand separately, gives it for d_∞. The triangle inequality for d_2 is the case $p = 2$, n finite of theorem 1.1.8 below. When $n = 1$ all three agree with $d(x, y) = |x - y|$, the metric that makes this book a generalization of the last one.

Example 1.1.4: The Sup Metric on $C[a, b]$

Let $C[a, b]$ denote the set of continuous functions $f\colon [a, b] \to \mathbb{R}$, and set

$$d_\infty(f, g) = \sup_{x\in[a,b]} |f(x) - g(x)|$$

The supremum is finite because a continuous function on a closed bounded interval is bounded [1], so $f - g$ is bounded. If $d_\infty(f, g) = 0$ then $f(x) = g(x)$ at every x, so $f = g$ as functions; symmetry is clear; and for the triangle inequality, for each fixed x,

$$|f(x) - h(x)| \leq |f(x) - g(x)| + |g(x) - h(x)| \leq d_\infty(f, g) + d_\infty(g, h)$$

so the right-hand side is an upper bound for the set whose supremum is $d_\infty(f, h)$.

A point of this space is an entire function, and two functions are close when their graphs lie in a thin horizontal band around one another. This is the space **??** is about, and d_∞ is the distance in which uniform convergence is ordinary convergence (**??**).

Example 1.1.5: The Discrete Metric

On any set X define $d(x, y) = 0$ if $x = y$ and $d(x, y) = 1$ otherwise. The first two axioms are immediate. For the triangle inequality, if $x = z$ the left side is 0; otherwise the left side is 1 and at least one of $d(x, y)$, $d(y, z)$ must be 1, since y cannot equal both x and z.

Every set carries a metric, then, so the existence of a metric says nothing about a set on its own. The discrete metric is the standard source of counterexamples: it is the extreme case in which no two distinct points are near each other, and it recurs below whenever a plausible-looking implication needs refuting.

The next example needs an inequality, and the inequality needs a smaller one. All three are stated for sequences, and specialize to finite sums by padding with zeros.

Proposition 1.1.6: Young's Inequality

Let $p, q > 1$ satisfy $\frac{1}{p} + \frac{1}{q} = 1$. Then for all real $a, b \geq 0$,

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}$$

***Proof*:**

If $a = 0$ or $b = 0$ the left side is 0 and the right side is nonnegative. Assume both are positive. The logarithm is concave, so for the weights $\frac{1}{p}$ and $\frac{1}{q}$, which are positive and sum to 1,

$$\log\left(\frac{a^p}{p} + \frac{b^q}{q}\right) \geq \frac{1}{p}\log(a^p) + \frac{1}{q}\log(b^q) = \log a + \log b = \log(ab)$$

Exponentiating, and using that the exponential is increasing, gives the claim, as we sought to show.

Theorem 1.1.7: Hölder's Inequality

Let $p, q > 1$ satisfy $\frac{1}{p} + \frac{1}{q} = 1$, and let $x = (x_k)_{k\geq 1}$ and $y = (y_k)_{k\geq 1}$ be real sequences. Then

$$\sum_{k=1}^{\infty} |x_k y_k| \leq \left(\sum_{k=1}^{\infty} |x_k|^p\right)^{1/p} \left(\sum_{k=1}^{\infty} |y_k|^q\right)^{1/q}$$

whenever the two factors on the right are finite.

***Proof*:**

Write $A = \left(\sum_k |x_k|^p\right)^{1/p}$ and $B = \left(\sum_k |y_k|^q\right)^{1/q}$. If $A = 0$ then every x_k is 0 and both sides vanish; likewise if $B = 0$. So assume $A, B > 0$ and set

$$a_k = \frac{|x_k|}{A} \qquad b_k = \frac{|y_k|}{B}$$

which satisfy $\sum_k a_k^p = 1$ and $\sum_k b_k^q = 1$. Applying proposition 1.1.6 termwise and summing,

$$\sum_{k=1}^{\infty} a_k b_k \leq \sum_{k=1}^{\infty} \left(\frac{a_k^p}{p} + \frac{b_k^q}{q}\right) = \frac{1}{p} + \frac{1}{q} = 1$$

Multiplying through by AB recovers the stated inequality, completing the proof.

Theorem 1.1.8: Minkowski's Inequality

Let $1 \leq p < \infty$ and let x, y be real sequences with $\sum_k |x_k|^p$ and $\sum_k |y_k|^p$ both finite. Then $\sum_k |x_k + y_k|^p$ is finite and

$$\left(\sum_{k=1}^{\infty} |x_k + y_k|^p\right)^{1/p} \leq \left(\sum_{k=1}^{\infty} |x_k|^p\right)^{1/p} + \left(\sum_{k=1}^{\infty} |y_k|^p\right)^{1/p}$$

***Proof*:**

Write $\|z\|_p = \left(\sum_k |z_k|^p\right)^{1/p}$ throughout.

Step 1: the left side is finite. For each k, $|x_k + y_k| \leq 2\max(|x_k|, |y_k|)$, so $|x_k + y_k|^p \leq 2^p\left(|x_k|^p + |y_k|^p\right)$, and summing shows $\|x + y\|_p < \infty$.

Step 2: the case $p = 1$. Here the inequality is the triangle inequality for $|\cdot|$ applied termwise and summed.

Step 3: the case $p > 1$. Let $q = \frac{p}{p-1}$, so that $\frac{1}{p} + \frac{1}{q} = 1$ and $(p-1)q = p$. Splitting one factor off each term,

$$\|x+y\|_p^p = \sum_k |x_k+y_k|\,|x_k+y_k|^{p-1} \le \sum_k |x_k|\,|x_k+y_k|^{p-1} + \sum_k |y_k|\,|x_k+y_k|^{p-1}$$

using $|x_k+y_k| \le |x_k| + |y_k|$. Apply theorem 1.1.7 to each sum on the right, pairing (x_k) or (y_k) with $\big(|x_k+y_k|^{p-1}\big)$:

$$\|x+y\|_p^p \le \big(\|x\|_p + \|y\|_p\big)\left(\sum_k |x_k+y_k|^{(p-1)q}\right)^{1/q} = \big(\|x\|_p + \|y\|_p\big)\|x+y\|_p^{p/q}$$

where the last equality uses $(p-1)q = p$. If $\|x+y\|_p = 0$ the claimed inequality is immediate, so assume it is positive and divide by $\|x+y\|_p^{p/q}$, which Step 1 guarantees is a finite positive number. Since $p - \frac{p}{q} = p\big(1-\frac{1}{q}\big) = p\cdot\frac{1}{p} = 1$, the left side becomes $\|x+y\|_p$, completing the proof.

Example 1.1.9: The Sequence Spaces ℓ^p

For $1 \le p < \infty$ let

$$\ell^p = \left\{ x = (x_k)_{k\ge 1} \;\middle|\; x_k \in \mathbb{R} \text{ and } \sum_{k=1}^{\infty} |x_k|^p < \infty \right\}$$

and let ℓ^∞ be the set of bounded real sequences. Define

$$d_p(x,y) = \left(\sum_{k=1}^{\infty} |x_k-y_k|^p\right)^{1/p} \qquad \text{and} \qquad d_\infty(x,y) = \sup_{k\ge 1} |x_k-y_k|$$

Theorem 1.1.8 shows first that $x-y \in \ell^p$ whenever $x,y \in \ell^p$, so d_p is defined and finite, and then, applied to $x-y$ and $y-z$, that

$$d_p(x,z) = \|(x-y)+(y-z)\|_p \le \|x-y\|_p + \|y-z\|_p = d_p(x,y) + d_p(y,z)$$

The other two axioms are immediate, as they are for d_∞, whose triangle inequality follows from the argument of example 1.1.3 with the maximum replaced by a supremum.

Concretely, the sequence $x = (1, \frac{1}{2}, \frac{1}{3}, \dots)$ lies in ℓ^2, since $\sum k^{-2}$ converges, but not in ℓ^1, since the harmonic series diverges, so the inclusion $\ell^1 \subseteq \ell^2$ is strict. Exercise 1.1.1 (3) establishes the whole chain $\ell^p \subsetneq \ell^r$ for $p < r$, with a witness for each strictness. The space ℓ^2 is one of this book's four running examples: it is the smallest space in which the difference between "closed and bounded" and "compact" becomes visible (**??**).

Every metric so far has been built from a notion of size that adds. The next one is built from divisibility, and it satisfies a triangle inequality much stronger than the one required.

Definition 1.1.10: Ultrametric

A metric d on X is an *ultrametric* if it satisfies the stronger inequality

$$d(x,z) \le \max\big(d(x,y),\, d(y,z)\big) \qquad \text{for all } x,y,z \in X$$

Example 1.1.11: The p-adic Metric on $\mathbb{Q}$

Fix a prime p. Every nonzero rational x factors uniquely as $x = p^v \frac{a}{b}$ with $v \in \mathbb{Z}$ and a, b nonzero integers coprime to p; write $v_p(x) = v$. Define

$$|x|_p = p^{-v_p(x)} \quad \text{for } x \neq 0, \qquad |0|_p = 0, \qquad d_p(x,y) = |x-y|_p$$

A rational is small in this metric when it is divisible by a high power of p. So $|3|_3 = \frac{1}{3}$ and $|\frac{1}{9}|_3 = 9$, and

$$d_3(1,10) = |-9|_3 = 3^{-2} = \tfrac{1}{9}$$

makes 1 and 10 closer than 1 and 2, whose distance is $|-1|_3 = 1$.

This is an ultrametric. For x, y nonzero with $x + y \neq 0$, writing $x = p^{v_p(x)}u$ and $y = p^{v_p(y)}w$ with u, w having no factor of p, the common factor $p^{\min(v_p(x), v_p(y))}$ can be extracted from $x + y$, so $v_p(x+y) \geq \min(v_p(x), v_p(y))$; negating the exponent reverses the inequality and gives $|x+y|_p \leq \max(|x|_p, |y|_p)$. Applying this to $x - z = (x-y) + (y-z)$ gives the ultrametric inequality, and the ordinary triangle inequality follows since a maximum is at most a sum of nonnegative terms. The arithmetic this metric opens up, and the field $\mathbb{Q}_p$ obtained by completing $\mathbb{Q}$ with respect to it (section 2.4), belong to *Local Field Theory* [2]; here it serves as an example of how little a metric needs to resemble $|x-y|$.

Proposition 1.1.12: Every Triangle in an Ultrametric Space Is Isosceles

Let (X, d) be an ultrametric space and let $x, y, z \in X$. If $d(x,y) \neq d(y,z)$ then $d(x,z) = \max\big(d(x,y), d(y,z)\big)$.

***Proof*:**

By symmetry of the hypothesis assume $d(x,y) < d(y,z)$, so the maximum is $d(y,z)$. The ultrametric inequality gives $d(x,z) \leq \max(d(x,y), d(y,z)) = d(y,z)$ directly. For the reverse, apply the ultrametric inequality to the pair (y,x), (x,z):

$$d(y,z) \leq \max\big(d(y,x),\, d(x,z)\big)$$

The first entry is $d(x,y)$, which is strictly less than $d(y,z)$, so it cannot be the maximum; hence the maximum is $d(x,z)$ and $d(y,z) \leq d(x,z)$. The two inequalities give equality, as we sought to show.

The conclusion is genuinely strange: in $\mathbb{Q}$ with d_3, no three points form a scalene triangle. Ultrametric spaces are the standard warning against reading metric statements through Euclidean pictures, and example 1.1.13 is a second such warning, this time inside the plane itself.

Example 1.1.13: The French Railway Metric

On $\mathbb{R}^2$, write $\|x\|$ for the Euclidean length and fix the origin O. Define

$$d(x,y) = \begin{cases} \|x-y\| & \text{if } x, y, O \text{ are collinear} \\ \|x\| + \|y\| & \text{otherwise} \end{cases}$$

Travel between two points is possible only along lines through the origin, so any journey not already radial must pass through O; the name records a rail network in which every route runs through the capital.

The first two axioms are clear. For the triangle inequality $d(x,z) \le d(x,y) + d(y,z)$, consider two cases. If x, z, O are collinear and y lies on that same line, all distances involved are Euclidean distances along a line and the inequality is the usual one. If x, z, O are collinear but y is not on their line, then no pair among $\{x, y\}$ and $\{y, z\}$ is radial, since y radial with x would put y on the line through O and x, which is the line through O and z. Hence

$$d(x,y) + d(y,z) = \|x\| + 2\|y\| + \|z\| \ge \|x\| + \|z\| \ge \|x - z\| = d(x,z)$$

the last step being the Euclidean triangle inequality applied to x, O, z. If instead x, z, O are not collinear then $d(x,z) = \|x\| + \|z\|$, and y cannot be radial with both x and z unless $y = O$, in which case $d(x,y) + d(y,z) = \|x\| + \|z\|$ with equality. Otherwise at least one of the two pairs is non-radial and contributes $\|x\| + \|y\|$ or $\|y\| + \|z\|$ in full, while the other contributes at least $\|z\| - \|y\|$ or $\|x\| - \|y\|$ by the Euclidean triangle inequality; adding gives at least $\|x\| + \|z\|$.

Exercise 1.1.1

1. Show that axiom (iii) of definition 1.1.1 cannot be dropped by exhibiting a function $d\colon X \times X \to \mathbb{R}_{\ge 0}$ on a three-point set satisfying (i) and (ii) but not (iii).
2. Let (X, d) be a metric space and define $d'(x,y) = \min\big(d(x,y), 1\big)$. Show that d' is a metric on X. Show also that $d''(x,y) = d(x,y)^2$ need not be one.
3. Show that $\ell^p \subseteq \ell^r$ whenever $1 \le p < r \le \infty$, and that the inclusion is strict. (For the inclusion, first reduce to the case $\|x\|_p = 1$ and observe what that forces about each $|x_k|$.)
4. In $\mathbb{Q}$ with the 5-adic metric, compute $d_5(1, 26)$, $d_5(\frac{1}{5}, 0)$ and $d_5(3, 3 + 5^4)$. Then show that the sequence $x_n = 5^n$ converges to 0, and that $x_n = n!$ does too.
5. Let (X, d) be an ultrametric space. Show that if $d(x,y) < r$ then every point z with $d(x,z) < r$ also satisfies $d(y,z) < r$; conclude that any two balls of radius r are either disjoint or equal, so that every point of a ball is a centre of it.
6. In the French railway metric of example 1.1.13, describe the set of points at distance less than r from a point $x \ne O$, distinguishing the cases $r \le \|x\|$ and $r > \|x\|$.

1.2 Open and Closed Sets

A metric assigns a number to each pair of points, which is more information than most arguments need. What they need is which points are near a given one, and that is captured by the balls: sets of the form “everything within r of x”. This section builds the standard vocabulary of nearness out of the balls alone, and in doing so isolates the part of a metric space that survives replacing the metric by a different but comparable one, a distinction section 1.4 makes precise.

Definition 1.2.1: Balls and Spheres

Let (X, d) be a metric space, $x \in X$ and $r > 0$. The *open ball* and *closed ball* of radius r about x are

$$B_r(x) = \{y \in X \mid d(x, y) < r\} \qquad \overline{B}_r(x) = \{y \in X \mid d(x, y) \leq r\}$$

and the *sphere* is $S_r(x) = \{y \in X \mid d(x, y) = r\}$. The point x is a *centre* of the ball and r a *radius*.

The indefinite article in "a centre" is deliberate. In $\mathbb{R}$ a ball determines its centre and radius, but exercise 1.1.1 (5) showed that in an ultrametric space every point of a ball is a centre of it, and in the discrete metric $B_2(x) = X$ for every x. Nothing below assumes a ball remembers how it was specified.

Definition 1.2.2: Open and Closed Sets

A subset $U \subseteq X$ is *open* if for every $x \in U$ there is an $r > 0$ with $B_r(x) \subseteq U$. A subset $A \subseteq X$ is *closed* if its complement $X \setminus A$ is open.

Open and closed are not opposites and not exhaustive. In $\mathbb{R}$ the interval $[0, 1)$ is neither, while $\emptyset$ and $\mathbb{R}$ are both. The definition is a condition at every point of the set, which is why the first thing to check is that the balls themselves satisfy it.

Proposition 1.2.3: Open Balls Are Open and Closed Balls Are Closed

Every open ball in a metric space is an open set, and every closed ball is a closed set.

***Proof*:**
Let $y \in B_r(x)$ and put $s = r - d(x, y)$, which is positive by the definition of the ball. If $d(y, z) < s$ then the triangle inequality gives

$$d(x, z) \leq d(x, y) + d(y, z) < d(x, y) + s = r$$

so $B_s(y) \subseteq B_r(x)$, which is the required condition at y.

For the closed ball, let $y \notin \overline{B}_r(x)$, so $d(x, y) > r$, and put $s = d(x, y) - r > 0$. If $d(y, z) < s$ then, rearranging the triangle inequality $d(x, y) \leq d(x, z) + d(z, y)$,

$$d(x, z) \geq d(x, y) - d(y, z) > d(x, y) - s = r$$

so $z \notin \overline{B}_r(x)$. Hence $B_s(y)$ misses $\overline{B}_r(x)$ entirely, the complement is open, and $\overline{B}_r(x)$ is closed, as we sought to show.

Theorem 1.2.4: The Metric Topology

Let (X, d) be a metric space and let $\mathcal{T}$ denote its collection of open sets. Then

(i) $\emptyset$ and X lie in $\mathcal{T}$;

(ii) any union of members of $\mathcal{T}$ lies in $\mathcal{T}$;

(iii) any *finite* intersection of members of $\mathcal{T}$ lies in $\mathcal{T}$.

***Proof*:**

(i) The condition on $\emptyset$ is vacuous, and for X any radius works since every ball is contained in X.

(ii) Let $\{U_\alpha\}_{\alpha\in A} \subseteq \mathcal{T}$ and let $x \in \bigcup_\alpha U_\alpha$. Then $x \in U_\beta$ for some β, and openness of U_β gives an $r > 0$ with $B_r(x) \subseteq U_\beta \subseteq \bigcup_\alpha U_\alpha$.

(iii) Let $U_1, \ldots, U_n \in \mathcal{T}$ and let $x \in \bigcap_i U_i$. Each U_i gives an $r_i > 0$ with $B_{r_i}(x) \subseteq U_i$. Put $r = \min_i r_i$, which is positive because the collection is finite. Then $B_r(x) \subseteq B_{r_i}(x) \subseteq U_i$ for every i, so $B_r(x) \subseteq \bigcap_i U_i$.

The finiteness in (iii) is not an artifact of the proof. In $\mathbb{R}$ the sets $B_{1/n}(0) = (-\frac{1}{n}, \frac{1}{n})$ are open for every n, but

$$\bigcap_{n=1}^{\infty} B_{1/n}(0) = \{0\}$$

which contains no ball at all. The failure is exactly that an infinite collection of positive radii can have infimum zero, which is what the proof's min avoids.

A collection satisfying the three conditions of theorem 1.2.4 is a *topology*, and the open sets of a metric space are its *metric topology*. This book needs the general notion only once, in **??**, and *Topology* [5] develops it properly; here the point of the theorem is that the balls do behave as the intervals of $\mathbb{R}$ do.

Definition 1.2.5: Convergent Sequence

A sequence $(x_n)_{n\geq 1}$ in a metric space X *converges* to $x \in X$ if for every $\epsilon > 0$ there is an N such that $d(x_n, x) < \epsilon$ for all $n \geq N$. Equivalently, every ball about x contains all but finitely many terms. Write $x_n \to x$, and call x a *limit* of the sequence.

Proposition 1.2.6: Limits Are Unique

A sequence in a metric space has at most one limit.

***Proof*:**
Suppose $x_n \to x$ and $x_n \to y$, and let $\epsilon > 0$. Choose N_1 with $d(x_n, x) < \epsilon/2$ for $n \geq N_1$ and N_2

with $d(x_n, y) < \epsilon/2$ for $n \geq N_2$. For any $n \geq \max(N_1, N_2)$ the triangle inequality gives

$$d(x, y) \leq d(x, x_n) + d(x_n, y) < \epsilon$$

Since $\epsilon > 0$ was arbitrary and $d(x, y)$ does not depend on it, $d(x, y) = 0$, and the first metric axiom forces $x = y$, completing the proof.

This is the promise made when the "only if" of definition 1.1.1 was insisted on. Under the pseudometric $d \equiv 0$ every sequence converges to every point, and the proof above breaks at exactly the last step: $d(x, y) = 0$ no longer says $x = y$.

Definition 1.2.7: Interior, Closure and Boundary

Let $A \subseteq X$. The *interior* A° is the union of all open sets contained in A; the *closure* $\overline{A}$ is the intersection of all closed sets containing A; the *boundary* is $\partial A = \overline{A} \setminus A^\circ$.

By theorem 1.2.4 the interior is open, being a union of open sets, and it is the largest open subset of A. The closure is closed, since an intersection of closed sets has open complement by de Morgan and part (ii) of the theorem, and it is the smallest closed superset of A. Neither description says which points belong, and for that the metric returns.

Definition 1.2.8: Limit Point

A point $x \in X$ is a *limit point* of $A \subseteq X$ if every ball about x meets $A \setminus \{x\}$. The set of limit points of A is written A'.

Theorem 1.2.9: Characterizations of the Closure

Let $A \subseteq X$ and $x \in X$. The following are equivalent:

(i) $x \in \overline{A}$;

(ii) every ball about x meets A;

(iii) x is the limit of some sequence of points of A.

Consequently $\overline{A} = A \cup A'$.

***Proof*:**
Step 1: (i) implies (ii). Suppose some ball $B_r(x)$ misses A. Then $X \setminus B_r(x)$ is a closed set containing A, so $\overline{A} \subseteq X \setminus B_r(x)$, and since $x \in B_r(x)$ this gives $x \notin \overline{A}$.

Step 2: (ii) implies (iii). For each $n \geq 1$ choose $a_n \in B_{1/n}(x) \cap A$, which is nonempty by hypothesis. Then $d(a_n, x) < 1/n$, so given $\epsilon > 0$ any N with $1/N < \epsilon$ works, and $a_n \to x$.

Step 3: (iii) implies (i). Let $a_n \to x$ with every $a_n \in A$, and let C be any closed set containing A. If $x \notin C$ then $X \setminus C$ is open and contains x, so it contains some $B_r(x)$, which therefore contains all a_n with n large. But those a_n lie in $A \subseteq C$, a contradiction. Hence x lies in every closed set containing A, that is, in $\overline{A}$.

For the last claim, if $x \in \overline{A}$ then by (ii) every ball about x meets A; either that intersection always contains a point other than x, making $x \in A'$, or some ball meets A only in x itself, forcing $x \in A$. Conversely $A \subseteq \overline{A}$ always, and a limit point of A satisfies (ii), completing the proof.

The third characterization is the one used most often, and it is worth naming the consequence for closed sets, since it converts a statement about complements into a statement about sequences.

Corollary 1.2.10: Closed Sets Contain Their Limits

A subset $A \subseteq X$ is closed if and only if $\overline{A} = A$, and if and only if the limit of every convergent sequence of points of A lies in A.

***Proof*:**
If A is closed then A is one of the closed sets containing A, so $\overline{A} \subseteq A$; the reverse inclusion always holds. If $\overline{A} = A$ then A is closed, being an intersection of closed sets. The equivalence with the sequential condition is theorem 1.2.9(iii): the limits of sequences in A are exactly the points of $\overline{A}$, and requiring them all to lie in A says exactly $\overline{A} \subseteq A$, as we sought to show.

Definition 1.2.11: Dense and Separable

A subset $D \subseteq X$ is *dense* if $\overline{D} = X$. The space X is *separable* if it has a countable dense subset.

By theorem 1.2.9, D is dense exactly when every ball in X contains a point of D: no point of the space can be isolated from D. Separability says that a countable list of points suffices to approximate every point of the space to any accuracy, and it is the hypothesis under which an uncountable space can still be handled by a sequence of choices.

Example 1.2.12: Separability of the Standard Spaces

(i) $\mathbb{R}^n$ is separable: the points with rational coordinates form a countable set, and given x and $\epsilon > 0$, choosing each coordinate within ϵ/n of x_i puts the resulting point q within ϵ of x in d_1. Since each of $d_\infty(x, q)$ and $d_2(x, q)$ is at most $d_1(x, q)$, the same q works for all three metrics of example 1.1.3.

(ii) ℓ^p is separable for $1 \leq p < \infty$. Take D to be the set of sequences with rational entries and only finitely many nonzero. Given $x \in \ell^p$ and $\epsilon > 0$, convergence of $\sum_k |x_k|^p$ gives an N with $\sum_{k>N} |x_k|^p < (\epsilon/2)^p$; truncating at N and then perturbing the first N entries to rationals within $\epsilon/(2N^{1/p})$ each puts a member of D within ϵ of x.

(iii) ℓ^∞ is *not* separable. For each subset $S \subseteq \mathbb{N}$ let χ_S be its indicator sequence. These are $2^{\aleph_0}$ many points of ℓ^∞, and $d_\infty(\chi_S, \chi_T) = 1$ whenever $S \neq T$, since the two differ by 1 at any index in exactly one of them. The balls $B_{1/2}(\chi_S)$ are therefore pairwise disjoint and uncountable in number, so any dense set must contain a point of each and cannot be countable.

That $C[0, 1]$ is separable is true but not elementary: the natural countable candidate is the set of polynomials with rational coefficients, and showing it is dense is the Weierstrass approximation theorem, proved in **??**.

The last example of this section is the fourth of the book's running spaces, and it is the standard source of a set that is closed, uncountable, and yet contains no ball whatsoever.

Example 1.2.13: The Cantor Set
Let $C_0 = [0, 1]$ and obtain C_{n+1} from C_n by deleting the open middle third of each of its constituent intervals, so that

$$C_1 = [0, \tfrac{1}{3}] \cup [\tfrac{2}{3}, 1] \qquad C_2 = [0, \tfrac{1}{9}] \cup [\tfrac{2}{9}, \tfrac{1}{3}] \cup [\tfrac{2}{3}, \tfrac{7}{9}] \cup [\tfrac{8}{9}, 1]$$

and C_n is a union of 2^n closed intervals of length 3^{-n}. The *Cantor set* is $C = \bigcap_{n \geq 0} C_n$, taken with the metric inherited from $\mathbb{R}$.

It is closed, being an intersection of closed sets. It has empty interior: any ball $B_r(x) \subseteq C$ would be an interval of positive length contained in every C_n, but the intervals making up C_n have length $3^{-n} \to 0$. And it is nonempty, since every endpoint of every interval of every C_n survives all later stages; indeed C consists exactly of the reals in $[0, 1]$ admitting a base-3 expansion using only the digits 0 and 2, which is why it is uncountable. Section 2.3 returns to it once completeness is available, where it becomes the standard example of a nonempty perfect set.

Exercise 1.2.1

1. Show that $\overline{B_r(x)} \subseteq \overline{B}_r(x)$ in any metric space, and give a metric space in which the inclusion is strict. (Try a space with very few points.)
2. Prove that $X \setminus A^\circ = \overline{X \setminus A}$ and $X \setminus \overline{A} = (X \setminus A)^\circ$ for every $A \subseteq X$.
3. Show that $\overline{A \cup B} = \overline{A} \cup \overline{B}$ for any two subsets, and that $\overline{A \cap B} \subseteq \overline{A} \cap \overline{B}$ with the inclusion possibly strict. Show also that the first identity can fail for infinite unions.
4. Describe the open sets, the closed sets and the convergent sequences of a set carrying the discrete metric of example 1.1.5. Which subsets are dense, and for which sets is the space separable?
5. Let X be separable and $Y \subseteq X$. Show that Y is separable in the inherited metric. (A countable dense subset of X need not lie in Y; for each pair (q, n) with q in the countable dense set, pick a point of $Y \cap B_{1/n}(q)$ when there is one.)
6. Show that the Cantor set contains no interval of positive length by computing the total length removed at all stages, and deduce that C has empty interior a second way.

1.3 Continuity in Three Forms

Single-variable analysis offers three descriptions of a continuous function: the ϵ-δ condition, the preservation of convergent sequences, and the condition that preimages of open sets be open. Each is stated using $|x - y|$ and nothing else, so each transcribes to a metric space unchanged. What is not automatic is that the three remain equivalent, and the first theorem of this section checks that they do.

The rest of the section separates continuity from two strengthenings that coincide with it on $\mathbb{R}$ often enough to be confused with it. Both differences matter later: uniform continuity is what **??** recovers for free on a compact domain, and the Lipschitz condition with constant less than 1 is the hypothesis of **??**.

Definition 1.3.1: Continuity

Let (X, d_X) and (Y, d_Y) be metric spaces. A function $f\colon X \to Y$ is *continuous at* $x \in X$ if for every $\epsilon > 0$ there is a $\delta > 0$ such that

$$d_X(x,t) < \delta \quad \Rightarrow \quad d_Y\big(f(x), f(t)\big) < \epsilon$$

and *continuous* if it is continuous at every point of X.

Theorem 1.3.2: Three Characterizations of Continuity

For $f\colon X \to Y$ between metric spaces the following are equivalent:

(i) f is continuous;

(ii) f preserves convergence: whenever $x_n \to x$ in X, $f(x_n) \to f(x)$ in Y;

(iii) $f^{-1}(V)$ is open in X for every open $V \subseteq Y$.

***Proof*:**
Step 1: (i) implies (ii). Let $x_n \to x$ and let $\epsilon > 0$. Choose δ as in definition 1.3.1 for this ϵ at the point x, and then N with $d_X(x_n, x) < \delta$ for $n \geq N$. For such n, $d_Y(f(x_n), f(x)) < \epsilon$.

Step 2: (ii) implies (i). Argue by contraposition. If f fails to be continuous at some x, there is an $\epsilon > 0$ for which no δ works; in particular $\delta = 1/n$ fails for each n, giving a point x_n with $d_X(x, x_n) < 1/n$ but $d_Y(f(x), f(x_n)) \geq \epsilon$. Then $x_n \to x$ while $f(x_n) \not\to f(x)$, so (ii) fails.

Step 3: (i) implies (iii). Let $V \subseteq Y$ be open and let $x \in f^{-1}(V)$. Openness of V gives an $\epsilon > 0$ with $B_\epsilon(f(x)) \subseteq V$, and continuity at x gives a $\delta > 0$ with $f\big(B_\delta(x)\big) \subseteq B_\epsilon(f(x)) \subseteq V$. Hence $B_\delta(x) \subseteq f^{-1}(V)$.

Step 4: (iii) implies (i). Fix $x \in X$ and $\epsilon > 0$. The ball $B_\epsilon(f(x))$ is open in Y by proposition 1.2.3, so $f^{-1}\big(B_\epsilon(f(x))\big)$ is open in X and contains x; it therefore contains some $B_\delta(x)$, and that δ satisfies the requirement of definition 1.3.1.

The third condition mentions no distances, only open sets, which is what makes it the definition topology adopts. The first two do mention distances, and the practical consequence is that a property defined through open sets alone cannot distinguish two metrics with the same open sets; section 1.4 takes this up, and chapter 2 exhibits a property that is *not* of this kind.

Proposition 1.3.3: Composition of Continuous Maps

If $f\colon X \to Y$ and $g\colon Y \to Z$ are continuous then so is $g \circ f$.

***Proof*:**
Let $W \subseteq Z$ be open. Then $(g \circ f)^{-1}(W) = f^{-1}\big(g^{-1}(W)\big)$, and $g^{-1}(W)$ is open by theorem 1.3.2(iii) applied to g, so its preimage under f is open by the same criterion applied to f. Hence $g \circ f$ is continuous, as we sought to show.

In definition 1.3.1 the number δ is allowed to depend on both ϵ and the point x. Forbidding the second dependence gives a strictly stronger condition, and forbidding it in a quantitative way gives a stronger one still.

Definition 1.3.4: Uniform Continuity

A function $f\colon X \to Y$ is *uniformly continuous* if for every $\epsilon > 0$ there is a $\delta > 0$, depending on ϵ alone, such that $d_X(x,t) < \delta$ implies $d_Y(f(x), f(t)) < \epsilon$ for *all* $x, t \in X$.

Definition 1.3.5: Lipschitz Map

A function $f\colon X \to Y$ is *Lipschitz* with constant $L \geq 0$ if

$$d_Y\big(f(x), f(t)\big) \leq L\, d_X(x,t) \qquad \text{for all } x, t \in X$$

A Lipschitz map with $L = 1$ is 1-*Lipschitz* or *nonexpanding*.

Proposition 1.3.6: The Continuity Hierarchy

Every Lipschitz map is uniformly continuous, and every uniformly continuous map is continuous.

***Proof*:**
Let f be Lipschitz with constant L and let $\epsilon > 0$. If $L = 0$ then f is constant and any δ serves; otherwise $\delta = \epsilon/L$ gives $d_Y(f(x), f(t)) \leq L\, d_X(x,t) < L\delta = \epsilon$, and δ depends only on ϵ. The second implication is immediate, since a δ valid at every point is in particular valid at each one, completing the proof.

Example 1.3.7: Separating the Three Notions

(i) $f(x) = x^2$ on $\mathbb{R}$ is continuous but not uniformly continuous. Given $\delta > 0$, take $x = 1/\delta$ and $t = x + \delta/2$; then $|x - t| < \delta$ while

$$|f(t) - f(x)| = |t - x|\,|t + x| = \frac{\delta}{2}\left(\frac{2}{\delta} + \frac{\delta}{2}\right) > 1$$

so no δ works for $\epsilon = 1$. The obstruction is that the function's steepness is unbounded, not any failure at a single point.

(ii) $f(x) = \sqrt{x}$ on $[0,1]$ is uniformly continuous but not Lipschitz. For uniform continuity, $|\sqrt{x} - \sqrt{t}|^2 \leq |\sqrt{x} - \sqrt{t}|\,|\sqrt{x} + \sqrt{t}| = |x - t|$, so $\delta = \epsilon^2$ works everywhere. It is not Lipschitz because

$$\frac{|f(x) - f(0)|}{|x - 0|} = \frac{\sqrt{x}}{x} = \frac{1}{\sqrt{x}} \to \infty \quad \text{as } x \to 0^+$$

so no single constant L bounds the ratio.

(iii) Any map out of a discrete metric space is Lipschitz with L equal to the diameter of its image when that is finite, since distinct points are at distance 1. Continuity is therefore no restriction at all on such a space.

Two maps built from the metric itself are Lipschitz, and both are used repeatedly. The first says the metric is continuous in its two arguments jointly.

Proposition 1.3.8: The Metric Is Continuous

For any metric space (X, d) and any $x, y, x', y' \in X$,

$$\left|d(x, y) - d(x', y')\right| \le d(x, x') + d(y, y')$$

Consequently, if $x_n \to x$ and $y_n \to y$ then $d(x_n, y_n) \to d(x, y)$.

***Proof*:**
Two applications of the triangle inequality give

$$d(x, y) \le d(x, x') + d(x', y') + d(y', y)$$

so $d(x, y) - d(x', y') \le d(x, x') + d(y, y')$ after using symmetry on the last term. Exchanging the roles of the primed and unprimed pairs gives the same bound for $d(x', y') - d(x, y)$, and the two together bound the absolute value. For the consequence, the right-hand side tends to 0, completing the proof.

Definition 1.3.9: Distance to a Set

For $\emptyset \ne A \subseteq X$ and $x \in X$, the *distance from x to A* is

$$d(x, A) = \inf_{a \in A} d(x, a)$$

Proposition 1.3.10: Distance to a Set Is 1-Lipschitz

Let $\emptyset \ne A \subseteq X$. Then $|d(x, A) - d(y, A)| \le d(x, y)$ for all $x, y \in X$, and

$$\overline{A} = \{x \in X \mid d(x, A) = 0\}$$

***Proof*:**
For any $a \in A$, the triangle inequality gives $d(x, A) \le d(x, a) \le d(x, y) + d(y, a)$. The left side does not involve a, so taking the infimum over $a \in A$ on the right yields $d(x, A) \le d(x, y) + d(y, A)$, that is, $d(x, A) - d(y, A) \le d(x, y)$. Exchanging x and y gives the other inequality, and the two bound the absolute value.

For the description of the closure, $d(x, A) = 0$ says that for every $\epsilon > 0$ some $a \in A$ has $d(x, a) < \epsilon$, that is, every ball about x meets A. By theorem 1.2.9(ii) this is exactly the condition $x \in \overline{A}$, as we sought to show.

Isomorphism in this subject comes in two strengths, and the gap between them is the subject of chapter 2.

Definition 1.3.11: Homeomorphism and Isometry

A bijection $f\colon X \to Y$ is a *homeomorphism* if both f and f^{-1} are continuous, and X and Y are then *homeomorphic*. It is an *isometry* if

$$d_Y\big(f(x), f(t)\big) = d_X(x, t) \qquad \text{for all } x, t \in X$$

and X and Y are then *isometric*.

An isometry is a homeomorphism: it is 1-Lipschitz, hence continuous by proposition 1.3.6, and its inverse satisfies the same identity, hence is continuous too. A homeomorphism preserves exactly the notions theorem 1.3.2(iii) can express, which is to say open sets, closures, convergence and continuity itself. An isometry preserves the metric outright, so it preserves in addition every quantity computed from distances: diameters, boundedness, and completeness.

Example 1.3.12: Homeomorphic but Not Isometric

The map $f\colon \mathbb{R} \to (-1, 1)$ given by

$$f(x) = \frac{x}{1 + |x|}$$

is a bijection with inverse $f^{-1}(y) = \frac{y}{1-|y|}$, and both are continuous as quotients of continuous functions with nonvanishing denominator, so $\mathbb{R}$ and $(-1, 1)$ are homeomorphic in their usual metrics. They are not isometric: $(-1, 1)$ has diameter 2 while $\mathbb{R}$ has infinite diameter, and an isometry preserves the set of distances realized.

The two spaces are therefore indistinguishable by every property expressible in open sets, yet one of them is complete and the other is not (section 2.1). Completeness is a property of the metric, not of the topology, and this pair is the standard example.

Exercise 1.3.1

1. Show that $f\colon X \to Y$ is continuous if and only if $f^{-1}(C)$ is closed in X for every closed $C \subseteq Y$. Show also that continuity does *not* imply that $f(U)$ is open for every open U, by considering a constant map.

2. Let $f, g\colon X \to \mathbb{R}$ be continuous. Show that $f + g$, $f - g$, fg and $\max(f, g)$ are continuous, and that f/g is continuous on $\{x \in X \mid g(x) \neq 0\}$. Deduce that every polynomial function $\mathbb{R}^n \to \mathbb{R}$ is continuous.

3. Show that $f(\overline{A}) \subseteq \overline{f(A)}$ for every continuous f and every $A \subseteq X$, and give an example where the inclusion is strict.

4. Call a sequence (x_n) *Cauchy* when for every $\epsilon > 0$ there is an N with $d(x_n, x_m) < \epsilon$ for all $n, m \geq N$. Show that a uniformly continuous map sends Cauchy sequences to Cauchy

sequences, and that a merely continuous map need not. (Use $f(x) = 1/x$ on $(0, 1)$.)

5. Let A, B be disjoint nonempty closed subsets of X. Show that

$$g(x) = \frac{d(x, A)}{d(x, A) + d(x, B)}$$

is a well-defined continuous function $X \to [0, 1]$ with $g \equiv 0$ on A and $g \equiv 1$ on B.

6. Show that every isometry is injective, and that a surjective 1-Lipschitz map need not be an isometry.

1.4 Subspaces, Products, and Equivalent Metrics

Three constructions appear in every later chapter, usually without comment. A subset of a metric space is a metric space; a product of two is a metric space; and a single space usually carries several metrics that one would like to treat as interchangeable. Each construction raises the same question, which is what survives it, and the third gives the language for answering all three: two metrics are interchangeable for topological purposes exactly when they have the same open sets, and interchangeable for metric purposes when each bounds a constant multiple of the other.

Definition 1.4.1: Subspace Metric

Let (X, d) be a metric space and $Y \subseteq X$. The *subspace metric* on Y is the restriction $d|_{Y \times Y}$, and Y with this metric is a *subspace* of X.

The axioms are inherited verbatim, so the only content is in how the open sets of Y relate to those of X. They are not the same sets, and the difference is a standard source of confusion.

Proposition 1.4.2: Open Sets of a Subspace

Let $Y \subseteq X$ carry the subspace metric. A set $U \subseteq Y$ is open in Y if and only if $U = V \cap Y$ for some V open in X. The balls satisfy $B_r^Y(y) = B_r^X(y) \cap Y$ for every $y \in Y$ and $r > 0$.

***Proof*:**
The statement about balls is immediate from the definition, since the condition $d(y, z) < r$ is the same whether z ranges over Y or over X.

Suppose $U = V \cap Y$ with V open in X, and let $y \in U$. Then $y \in V$, so some $B_r^X(y) \subseteq V$, whence $B_r^Y(y) = B_r^X(y) \cap Y \subseteq V \cap Y = U$. So U is open in Y.

Conversely let U be open in Y. For each $y \in U$ choose $r_y > 0$ with $B_{r_y}^Y(y) \subseteq U$, and put $V = \bigcup_{y \in U} B_{r_y}^X(y)$, which is open in X by theorem 1.2.4. Then

$$V \cap Y = \bigcup_{y \in U} \left(B_{r_y}^X(y) \cap Y\right) = \bigcup_{y \in U} B_{r_y}^Y(y)$$

and this union contains every $y \in U$ while being contained in U, so it equals U, as we sought to show.

So $[0, \frac{1}{2})$ is open in the subspace $[0, 1]$ of $\mathbb{R}$, being $(-\frac{1}{2}, \frac{1}{2}) \cap [0, 1]$, though it is not open in $\mathbb{R}$. Openness is always relative to an ambient space, and in this book the ambient space is named whenever it is not the obvious one.

Definition 1.4.3: Product Metrics

Let (X, d_X) and (Y, d_Y) be metric spaces. For $m = (x, y)$ and $m' = (x', y')$ in $X \times Y$ define

$$\begin{aligned} d_E(m, m') &= \sqrt{d_X(x, x')^2 + d_Y(y, y')^2} \\ d_{\max}(m, m') &= \max\left\{d_X(x, x'),\, d_Y(y, y')\right\} \\ d_{\text{sum}}(m, m') &= d_X(x, x') + d_Y(y, y') \end{aligned}$$

Each is a metric. For d_{sum} and $d_{\max}$ the verification is the coordinatewise argument of example 1.1.3; for d_E it is the case $p = 2$, $n = 2$ of theorem 1.1.8 applied to the pair of nonnegative numbers $\big(d_X(x, x'), d_Y(y, y')\big)$ and $\big(d_X(x', x''), d_Y(y', y'')\big)$, together with the fact that each coordinate distance already obeys a triangle inequality and $t \mapsto t^2$ is increasing on $[0, \infty)$.

Rather than choose between the three, the next result shows the choice does not matter, in a sense the rest of the section makes precise.

Proposition 1.4.4: Comparison of the Product Metrics

On $X \times Y$,

$$d_{\max} \leq d_E \leq d_{\text{sum}} \leq 2\, d_{\max}$$

pointwise.

***Proof*:**
Write $a = d_X(x, x')$ and $b = d_Y(y, y')$, both nonnegative.

For the first inequality, $\max(a, b)^2$ is either a^2 or b^2, and each is at most $a^2 + b^2$; taking square roots, which preserves order on $[0, \infty)$, gives $\max(a, b) \leq \sqrt{a^2 + b^2}$.

For the second, $(a + b)^2 = a^2 + 2ab + b^2 \geq a^2 + b^2$ since $ab \geq 0$, and taking square roots gives $\sqrt{a^2 + b^2} \leq a + b$.

For the third, both a and b are at most $\max(a, b)$, so their sum is at most $2 \max(a, b)$, completing the proof.

Definition 1.4.5: Equivalent Metrics

Two metrics d and d' on the same set X are *topologically equivalent* if they have the same open sets. They are *strongly equivalent* if there are constants $c_1, c_2 > 0$ with

$$c_1\, d(x, y) \leq d'(x, y) \leq c_2\, d(x, y) \qquad \text{for all } x, y \in X$$

Proposition 1.4.6: Strong Equivalence Implies Topological

Strongly equivalent metrics are topologically equivalent.

***Proof*:**
Let U be open for d and let $x \in U$, so $B_r^d(x) \subseteq U$ for some $r > 0$. If $d'(x, z) < c_1 r$ then $c_1 d(x, z) \leq d'(x, z) < c_1 r$, so $d(x, z) < r$ and $z \in U$. Hence $B_{c_1 r}^{d'}(x) \subseteq U$ and U is open for d'. The reverse inclusion follows by the symmetric argument using c_2, completing the proof.

Corollary 1.4.7: The Three Product Metrics Are Interchangeable

The metrics d_E, $d_{\max}$ and d_{sum} on $X \times Y$ are strongly equivalent, hence topologically equivalent. On $\mathbb{R}^n$ the three metrics of example 1.1.3 satisfy

$$d_\infty \leq d_2 \leq d_1 \leq n\, d_\infty$$

and are therefore strongly equivalent as well.

***Proof*:**
Proposition 1.4.4 gives $d_{\max} \leq d_E \leq d_{\text{sum}} \leq 2d_{\max}$, so each pair among the three is bracketed by constant multiples of the other, and proposition 1.4.6 applies. On $\mathbb{R}^n$ the same argument with n coordinates in place of two gives $d_\infty \leq d_2 \leq d_1 \leq n\, d_\infty$, completing the proof.

Whenever $\mathbb{R}^n$ or a finite product appears below, no metric is specified: by corollary 1.4.7 every statement about open sets, convergence, continuity or closure is independent of the choice.

The converse of proposition 1.4.6 fails, and the failure is the reason the two notions have separate names.

Example 1.4.8: Topologically but Not Strongly Equivalent
On $\mathbb{R}$ let $d(x, y) = |x - y|$ and $d'(x, y) = \min\big(|x - y|, 1\big)$, which is a metric by exercise 1.1.1 (2). They are topologically equivalent: any d'-ball of radius $r \leq 1$ is the d-ball of the same radius, and balls of radius at most 1 already witness openness for both metrics.

They are not strongly equivalent. Strong equivalence would give a $c_1 > 0$ with $c_1|x - y| \leq \min(|x - y|, 1)$ for all x, y, and taking $y = 0$ and $x \to \infty$ makes the left side unbounded while the right side never exceeds 1.

The two metrics therefore have identical open sets, identical convergent sequences and identical continuous functions, yet differ on every question involving actual distances: $\mathbb{R}$ has infinite diameter under d and diameter 1 under d', and a set can be d'-bounded without being d-bounded.

Proposition 1.4.9: Convergence in a Product Is Coordinatewise

A sequence $\big((x_n, y_n)\big)$ in $X \times Y$ converges to (x, y) for any one of the three product metrics if and only if $x_n \to x$ in X and $y_n \to y$ in Y.

***Proof*:**
By corollary 1.4.7 it suffices to argue for $d_{\max}$. If $d_{\max}\big((x_n, y_n), (x, y)\big) < \epsilon$ then both $d_X(x_n, x) < \epsilon$ and $d_Y(y_n, y) < \epsilon$, since each is at most the maximum, so convergence in the product forces convergence in each coordinate. Conversely, given $\epsilon > 0$ choose N_1 and N_2 beyond which the two coordinate distances are less than ϵ; for $n \geq \max(N_1, N_2)$ the maximum is less than ϵ, as

we sought to show.

Products of finitely many factors are handled by any of the three formulas. For countably many factors all three fail: a supremum of infinitely many distances can be infinite, and a sum of infinitely many can diverge. Rescaling each factor to have diameter at most 1 and then damping the terms repairs both problems at once.

Definition 1.4.10: Countable Product Metric

Let $(X_n, d_n)_{n\geq 1}$ be metric spaces. On $\prod_{n\geq 1} X_n$ define

$$d(x,y) = \sum_{n=1}^{\infty} 2^{-n} \min\left(d_n(x_n,y_n),\, 1\right)$$

Proposition 1.4.11: The Countable Product Metric

The function d of definition 1.4.10 is a metric, and $x^{(k)} \to x$ with respect to it if and only if $x_n^{(k)} \to x_n$ in X_n for every n.

***Proof*:**
The series converges, being dominated termwise by $\sum 2^{-n} = 1$. Each summand is a metric on X_n by exercise 1.1.1 (2), so symmetry and the triangle inequality hold termwise and are preserved by the (convergent) sum. If $d(x,y) = 0$ then every summand vanishes, forcing $x_n = y_n$ for all n.

For the convergence claim, suppose first $d(x^{(k)}, x) \to 0$ and fix n. Since the nth summand is at most the whole sum,

$$2^{-n} \min\left(d_n(x_n^{(k)}, x_n), 1\right) \leq d(x^{(k)}, x) \to 0$$

so $\min(d_n(x_n^{(k)}, x_n), 1) \to 0$, which forces $d_n(x_n^{(k)}, x_n) \to 0$ because the minimum equals the distance once the distance drops below 1.

Conversely suppose every coordinate converges and let $\epsilon > 0$. Choose N with $2^{-N} < \epsilon/2$, so the tail $\sum_{n>N} 2^{-n} \min(\cdot, 1) \leq \sum_{n>N} 2^{-n} = 2^{-N} < \epsilon/2$ regardless of the points involved. For each of the finitely many $n \leq N$, choose K_n with $d_n(x_n^{(k)}, x_n) < \epsilon/2$ for $k \geq K_n$. For $k \geq \max_{n\leq N} K_n$,

$$d(x^{(k)}, x) < \sum_{n=1}^{N} 2^{-n} \frac{\epsilon}{2} + \frac{\epsilon}{2} < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

completing the proof.

The damping factors 2^{-n} are one choice among many; any positive summable sequence produces a metric with the same convergent sequences. What the construction shows is that countable products stay metrizable, which finite products achieve more cheaply and uncountable products do not achieve at all.

Example 1.4.12: The Hilbert Cube

Taking every factor to be $[0,1]$ with its usual metric gives the *Hilbert cube* $[0,1]^{\mathbb{N}}$, whose points are sequences in $[0,1]$ and whose metric is

$$d(x,y) = \sum_{n=1}^{\infty} 2^{-n}|x_n - y_n|$$

the minimum being unnecessary since all distances are already at most 1. By proposition 1.4.11 convergence here is coordinatewise, so the sequence $x^{(k)} = (0,\ldots,0,1,0,\ldots)$ with the 1 in position k converges to the zero sequence, each coordinate being eventually 0. In ℓ^∞, by contrast, these same points are pairwise at distance 1 (example 1.2.12), so they converge to nothing. Coordinatewise convergence and uniform convergence are different notions, and **??** is about the difference.

Exercise 1.4.1

1. Let $Y \subseteq X$ be a subspace. Show that $A \subseteq Y$ is closed in Y if and only if $A = C \cap Y$ for some C closed in X. Show that if Y is itself closed in X, then a set closed in Y is closed in X, and give an example showing this fails without that hypothesis.
2. Show that the projections $\pi_X \colon X \times Y \to X$ and $\pi_Y \colon X \times Y \to Y$ are 1-Lipschitz for $d_{\max}$, and that a map $f \colon Z \to X \times Y$ is continuous if and only if $\pi_X \circ f$ and $\pi_Y \circ f$ both are.
3. Show that strongly equivalent metrics have the same Cauchy sequences and the same bounded sets, and that topologically equivalent metrics need not, using example 1.4.8.
4. Verify that topological equivalence and strong equivalence are each equivalence relations on the set of metrics on a fixed set X.
5. Let X be an infinite set with the discrete metric d, and let $d'(x,y) = \min(d(x,y), \frac{1}{2})$ for $x \neq y$. Determine whether d and d' are topologically equivalent, and whether they are strongly equivalent.
6. Compute the diameter of the Hilbert cube under the metric of example 1.4.12, and exhibit two points realizing it.

2

Completeness

Convergence, as definition 1.2.5 states it, requires a limit to be named in advance. That is an awkward requirement, because the sequences analysis produces are usually built by a procedure, the procedure produces terms rather than a limit, and whether a limit exists is the question being asked. What is available instead is the observation that the terms crowd together, and the whole content of this chapter is that in the right spaces this observation is enough.

Spaces where it is enough are the complete ones, and the property turns out to be sharper than it first appears. Section 2.1 defines it and shows that it is a property of the metric and not of the open sets: two spaces can be homeomorphic with one complete and the other not, so no argument phrased in open sets alone can establish it. Section 2.2 verifies it for the spaces this book actually uses, and the verification for $C_b(X)$ is the one **??** rests on. Section 2.3 extracts what completeness gives, in the form of a nested-set principle that produces points nobody has constructed, and uses it to show that a nonempty complete space with no isolated points is uncountable. Section 2.4 completes the chapter by showing that an incomplete space can always be repaired, in exactly one way.

2.1 Cauchy Sequences and Complete Spaces

Convergence, as definition 1.2.5 states it, is a relation between a sequence and a point. This section isolates the condition that remains when the point is struck out of the statement, and names the spaces in which the two conditions agree.

To test a sequence against definition 1.2.5 one must already hold the limit in hand, since the limit occurs in the condition being tested. That is a real restriction. A sequence produced by an iteration, an algorithm or a series comes with no candidate limit attached, and whether such a point exists is exactly the question at issue. What can be tested without a candidate is whether the terms of the sequence eventually crowd together.

Definition 2.1.1: Cauchy Sequence

A sequence $(x_n)_{n\geq 1}$ in a metric space (X, d) is *Cauchy* if for every $\epsilon > 0$ there is an N such that

$$d(x_n, x_m) < \epsilon \qquad \text{for all } n, m \geq N$$

The difference from definition 1.2.5 is one symbol. Convergence bounds $d(x_n, x)$, where x is a point of X named in advance; the Cauchy condition bounds $d(x_n, x_m)$, where both entries are terms of the sequence. No point outside the sequence appears in definition 2.1.1, so the condition can be checked by someone who knows the terms and nothing else about the space.

A computation makes the difference visible. In $\mathbb{R}$ let $x_n = \sum_{k=1}^{n} 1/k^2$. For $m > n$ the terms $1/k^2$ with $k \geq 2$ satisfy $1/k^2 \leq 1/\big(k(k-1)\big)$, and that bound telescopes:

$$|x_m - x_n| = \sum_{k=n+1}^{m} \frac{1}{k^2} \leq \sum_{k=n+1}^{m} \frac{1}{k(k-1)} = \sum_{k=n+1}^{m} \left(\frac{1}{k-1} - \frac{1}{k}\right) = \frac{1}{n} - \frac{1}{m} < \frac{1}{n}$$

So given $\epsilon > 0$, any $N > 1/\epsilon$ forces $|x_m - x_n| < 1/N < \epsilon$ for all $m > n \geq N$, and the case $m = n$ is trivial. The sequence is Cauchy. Nothing in the computation refers to a limit, and nothing in it produces one: the value $\pi^2/6$ is invisible to the estimate.

The two conditions are not independent. Bounding the distance from every term to a common point bounds the distances between terms, by the triangle inequality, and that is the whole of one implication.

Proposition 2.1.2: Convergent Sequences Are Cauchy

Every convergent sequence in a metric space is Cauchy.

***Proof*:**
Let $x_n \to x$ in (X, d) and let $\epsilon > 0$. By definition 1.2.5 there is an N with $d(x_n, x) < \epsilon/2$ for all $n \geq N$. For any $n, m \geq N$, the triangle inequality and symmetry give

$$d(x_n, x_m) \leq d(x_n, x) + d(x, x_m) < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

as we sought to show.

The converse is false, and each failure of it is a failure of the space rather than of the sequence: the terms crowd together and the space gives no point for them to crowd around. Two properties hold of every Cauchy sequence regardless, and both are proved without a limit. The first is the metric form of the statement that a convergent real sequence is bounded.

Proposition 2.1.3: Cauchy Sequences Are Bounded

Let $(x_n)_{n\geq 1}$ be a Cauchy sequence in a metric space X. Then there is an $R > 0$ with $x_n \in B_R(x_1)$ for every n; that is, the set of terms $\{x_n \mid n \geq 1\}$ is bounded.

***Proof*:**
Apply definition 2.1.1 with $\epsilon = 1$: there is an N with $d(x_n, x_m) < 1$ for all $n, m \geq N$. Put

$$R = 1 + \max_{1 \leq j \leq N} d(x_j, x_1)$$

which is a maximum over a finite nonempty index set, hence a real number, and which satisfies $R \geq 1 > 0$ because the maximum is at least $d(x_1, x_1) = 0$.

For $n \leq N$ the term $d(x_n, x_1)$ is one of those entering the maximum, so $d(x_n, x_1) \leq R - 1 < R$. For $n > N$ both indices n and N are at least N, so the Cauchy estimate applies to the pair and

$$d(x_n, x_1) \leq d(x_n, x_N) + d(x_N, x_1) < 1 + d(x_N, x_1) \leq R$$

In either case $x_n \in B_R(x_1)$, completing the proof.

Boundedness is necessary and not sufficient. In $\mathbb{R}$ the sequence $x_n = (-1)^n$ has all its terms in $B_2(0)$, yet $|x_n - x_{n+1}| = 2$ for every n, so no N meets the requirement of definition 2.1.1 for $\epsilon = 1$. The Cauchy condition asks the terms to accumulate at a single location, and boundedness only confines them to a bounded region.

The second property matters whenever the limit is obtained by extraction rather than built outright. Some completeness proofs construct the candidate limit directly, coordinate by coordinate or pointwise. Others, and in particular those routed through compactness, produce a limit only for a subsequence obtained by a construction such as repeated bisection of an interval, or a diagonal choice across a countable family of sequences; **??** carries out both. Extraction needs a name.

Definition 2.1.4: Subsequence

A *subsequence* of a sequence $(x_n)_{n\geq 1}$ in X is a sequence $(x_{n_k})_{k\geq 1}$ determined by a strictly increasing choice of indices $n_1 < n_2 < n_3 < \cdots$ in $\mathbb{N}$.

Such a choice satisfies $n_k \geq k$ for every k: the indices are positive integers, so $n_1 \geq 1$, and $n_{k+1} > n_k \geq k$ forces $n_{k+1} \geq k + 1$. That inequality is what lets an estimate valid beyond index N be applied to the kth term of any subsequence, and it is used without comment below. For a Cauchy sequence the extraction loses nothing.

Proposition 2.1.5: A Cauchy Sequence with a Convergent Subsequence Converges

Let $(x_n)_{n\geq 1}$ be a Cauchy sequence in a metric space X, and suppose some subsequence $(x_{n_k})_{k\geq 1}$ converges to $x \in X$. Then $x_n \to x$.

***Proof*:**
Let $\epsilon > 0$. Since (x_n) is Cauchy there is an N with $d(x_n, x_m) < \epsilon/2$ for all $n, m \geq N$, and since $x_{n_k} \to x$ there is a K with $d(x_{n_k}, x) < \epsilon/2$ for all $k \geq K$.

Fix one index $k \geq \max(K, N)$. Then $n_k \geq k \geq N$, so for every $n \geq N$ the pair (n, n_k) falls under the Cauchy estimate, and the triangle inequality gives

$$d(x_n, x) \leq d(x_n, x_{n_k}) + d(x_{n_k}, x) < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

The index k was fixed after ϵ, and the resulting bound holds for all $n \geq N$ with N depending on ϵ alone. Hence $x_n \to x$, as we sought to show.

Neither hypothesis can be dropped. A Cauchy sequence need not converge, as the examples below show, and a sequence with a convergent subsequence need not converge either: $x_n = (-1)^n$ in $\mathbb{R}$ has the constant subsequence $x_{2k} = 1$ converging to 1, while (x_n) itself converges to nothing, since a convergent sequence is Cauchy by proposition 2.1.2 and this one was seen above not to be. It is the conjunction that forces convergence. **??** produces convergent subsequences from hypotheses of an entirely different kind, and proposition 2.1.5 is what converts each such extraction into a statement about the original sequence.

For sequences of real numbers the Cauchy condition and convergence agree, and that agreement is one of the standard formulations of the completeness of $\mathbb{R}$ [1]. In a general metric space it is not a theorem but a hypothesis.

Definition 2.1.6: Complete Metric Space

A metric space (X, d) is *complete* if every Cauchy sequence in X converges to a point of X. A subset $A \subseteq X$ is *complete* if it is complete as a metric space under the subspace metric of definition 1.4.1.

By proposition 2.1.2 the convergent sequences of a metric space always sit among its Cauchy sequences, so completeness is the assertion that the two classes coincide. The content is an existence statement: a description of a point by successive approximations, given without naming the point, always picks out an actual point of the space. Failure of the property is therefore always the same phenomenon, a description with no object, and the two standard examples exhibit it by deleting the object from a space that had it.

Example 2.1.7: Two Incomplete Spaces

(i) $\mathbb{Q}$ with the metric $d(x,y) = |x-y|$. Define a sequence of rationals by

$$x_1 = 2 \qquad x_{n+1} = \frac{x_n}{2} + \frac{1}{x_n} = \frac{x_n^2+2}{2x_n}$$

Each x_n is a positive rational: this holds for x_1, and if x_n is a positive rational then so is $(x_n^2+2)/(2x_n)$. The first four terms are 2, $\frac{3}{2}$, $\frac{17}{12}$ and $\frac{577}{408}$.

Write $e_n = x_n^2 - 2$. Squaring the recursion,

$$e_{n+1} = \frac{(x_n^2+2)^2 - 8x_n^2}{4x_n^2} = \frac{x_n^4 - 4x_n^2 + 4}{4x_n^2} = \frac{(x_n^2-2)^2}{4x_n^2} = \frac{e_n^2}{4x_n^2}$$

so $e_n \geq 0$ for every $n \geq 2$, and $e_1 = 2 \geq 0$ as well. Hence $x_n^2 \geq 2$ for all n, and since each x_n is positive this gives $x_n > 1$ and $4x_n^2 \geq 8$. Therefore $e_{n+1} \leq e_n^2/8$, and induction gives $e_n \leq 2$ for all n: it holds at $n = 1$, and $e_n \leq 2$ forces $e_{n+1} \leq 4/8 < 2$. Feeding $e_n \leq 2$ back in,

$$e_{n+1} \leq \frac{e_n \cdot e_n}{8} \leq \frac{2e_n}{8} = \frac{e_n}{4}$$

so $e_n \leq 2 \cdot 4^{1-n}$ for every n.

The sequence decreases, since $x_n - x_{n+1} = \frac{x_n}{2} - \frac{1}{x_n} = \frac{e_n}{2x_n} \geq 0$. So for $m \geq n$ both $x_n \geq x_m$ and $x_n + x_m > 2$ hold, and $x_m^2 \geq 2$ gives

$$0 \leq x_n - x_m = \frac{x_n^2 - x_m^2}{x_n + x_m} \leq \frac{x_n^2 - 2}{x_n + x_m} \leq \frac{e_n}{2} \leq 4^{1-n}$$

Given $\epsilon > 0$, choosing N with $4^{1-N} < \epsilon$ makes $|x_n - x_m| < \epsilon$ for all $n, m \geq N$, so (x_n) is Cauchy in $\mathbb{Q}$.

It has no limit in $\mathbb{Q}$. Suppose $x_n \to q$ with $q \in \mathbb{Q}$. Every x_n lies in $(1, 2]$, so

$$|q^2 - 2| \leq |q^2 - x_n^2| + |x_n^2 - 2| = |q - x_n|\,|q + x_n| + e_n \leq \big(|q| + 2\big)|q - x_n| + 2 \cdot 4^{1-n}$$

Both terms on the right tend to 0, and the left side does not depend on n, so $q^2 = 2$. No rational has square 2: writing $q = a/b$ in lowest terms, $a^2 = 2b^2$ makes a even, say $a = 2c$, whence $b^2 = 2c^2$ makes b even, contradicting the choice of lowest terms. So $\mathbb{Q}$ is not complete.

Inside $\mathbb{R}$ the same sequence converges, to $\sqrt{2}$. The terms are not defective; the space is missing the point they approximate.

(ii) The interval $(0, 1)$ with the metric inherited from $\mathbb{R}$, and $x_n = 1/n$ for $n \geq 2$. For $m \geq n \geq N$,

$$|x_n - x_m| = \left|\frac{1}{n} - \frac{1}{m}\right| \leq \frac{1}{n} \leq \frac{1}{N}$$

so any $N > 1/\epsilon$ witnesses the Cauchy condition for ϵ. Suppose $1/n \to x$ with $x \in (0, 1)$. Then $x > 0$, and every $n \geq 2/x$ satisfies $1/n \leq x/2$, hence

$$|x_n - x| = x - \frac{1}{n} \geq x - \frac{x}{2} = \frac{x}{2}$$

So no N works for $\epsilon = x/2$, and the sequence has no limit in $(0, 1)$.

Both failures have the same shape. Each space sits inside a larger one where the sequence does converge, and each fails to contain the limit: $\sqrt{2} \notin \mathbb{Q}$ and $0 \notin (0, 1)$. In both cases the offending subset is not closed in the larger space, $\overline{\mathbb{Q}} = \mathbb{R}$ and $\overline{(0, 1)} = [0, 1]$, and that observation is the whole content of the next theorem, which turns completeness of a subspace into a question about closure.

Theorem 2.1.8: Closed Subspaces Are Complete and Complete Subspaces Are Closed

Let (X, d) be a metric space and let $A \subseteq X$ carry the subspace metric.

(i) If X is complete and A is closed in X, then A is complete.

(ii) If A is complete, then A is closed in X, whether or not X is complete.

***Proof*:**
Throughout, the subspace metric on A is the restriction of d (definition 1.4.1), so for points of A the quantity $d(a, a')$ is the same computed in A or in X. A sequence in A is therefore Cauchy in A if and only if it is Cauchy in X, and it converges in A to a point $a \in A$ if and only if it

converges to a in X.

(i) Let (a_n) be a Cauchy sequence in A. By the previous paragraph it is Cauchy in X, so completeness of X gives an $x \in X$ with $a_n \to x$. Every term lies in A and A is closed, so $x \in A$ by corollary 1.2.10. Convergence of (a_n) to a point of A in X is convergence in A, so (a_n) converges in A. Hence A is complete.

(ii) By corollary 1.2.10 it suffices to show that the limit of every convergent sequence of points of A lies in A. So let $a_n \in A$ with $a_n \to x$ in X. The sequence is convergent in X, hence Cauchy in X by proposition 2.1.2, hence Cauchy in A. Completeness of A gives an $a \in A$ with $a_n \to a$ in A, and therefore $a_n \to a$ in X as well. A sequence in X has at most one limit by proposition 1.2.6, so $x = a \in A$, completing the proof.

The two halves have unequal hypotheses. Statement (ii) asks nothing of the ambient space: a complete subset of a metric space is closed in it, whether or not the ambient space is itself complete. Statement (i) does need X complete, and without that hypothesis it fails at once: $\mathbb{Q}$ is closed in $\mathbb{Q}$ and is not complete.

Inside a complete space the two conditions collapse into one, and that collapse is how completeness gets checked for every subspace of a space already known to be complete. The Cantor set C of example 1.2.13 is an intersection of closed subsets of $\mathbb{R}$, hence closed, hence complete. The interval $[0, 1]$ is complete for the same reason, while $(0, 1)$ is not closed in $\mathbb{R}$ and example 2.1.7 confirms that it is not complete. The criterion also recovers the first example: $\mathbb{Q}$ is dense in $\mathbb{R}$ (the case $n = 1$ of example 1.2.12), so $\overline{\mathbb{Q}} = \mathbb{R} \neq \mathbb{Q}$ and $\mathbb{Q}$ is not closed.

That criterion is stated relative to an ambient space, and it invites the question of whether completeness might, like openness or convergence, be a matter of the open sets alone. It is not, and the following example is the standard witness.

Example 2.1.9: Completeness Is Not Preserved by Homeomorphism

The map $f\colon \mathbb{R} \to (-1, 1)$ of example 1.3.12, given by $f(x) = x/(1 + |x|)$, is a homeomorphism. The line $\mathbb{R}$ is complete [1]. The interval $(-1, 1)$ is not: the points $x_n = 1 - 1/n$ all lie in $(-1, 1)$, and for $m \geq n \geq N$

$$|x_n - x_m| = \left| \frac{1}{m} - \frac{1}{n} \right| \leq \frac{1}{N}$$

so taking $N > 1/\epsilon$ makes (x_n) Cauchy. A limit $x \in (-1, 1)$ would also be a limit in $\mathbb{R}$, where the sequence converges to 1, so uniqueness of limits (proposition 1.2.6) would force $x = 1 \notin (-1, 1)$.

So two homeomorphic spaces differ on completeness. The difference can be arranged without changing the underlying set at all. On $\mathbb{R}$ define

$$\rho(x, y) = |f(x) - f(y)|$$

This is a metric: symmetry and the triangle inequality are inherited from the metric of $\mathbb{R}$ applied to the values $f(x)$, and $\rho(x, y) = 0$ gives $f(x) = f(y)$, hence $x = y$ since f is injective.

The metrics ρ and $d(x, y) = |x - y|$ are topologically equivalent in the sense of definition 1.4.5. Regarded as a map $(\mathbb{R}, \rho) \to (-1, 1)$, f is a surjective isometry by the definition of ρ, hence a homeomorphism (definition 1.3.11); regarded as a map $(\mathbb{R}, d) \to (-1, 1)$ it is a homeomorphism by example 1.3.12. The identity map $(\mathbb{R}, d) \to (\mathbb{R}, \rho)$ is the composite of the second with the inverse of the first, hence a homeomorphism by proposition 1.3.3, and a homeomorphism which is the identity on the underlying set says exactly that the two metrics have the same open sets.

Yet $(\mathbb{R}, \rho)$ is not complete. Take $x_n = n$. Then

$$\rho(n,m) = \left|\frac{n}{1+n} - \frac{m}{1+m}\right| = \frac{|n-m|}{(1+n)(1+m)} \leq \frac{1}{1+\min(n,m)}$$

where the last step uses $|n-m| \leq \max(n,m) < 1+\max(n,m)$ together with the factorization $(1+n)(1+m) = \big(1+\max(n,m)\big)\big(1+\min(n,m)\big)$. Given $\epsilon > 0$, any $N > 1/\epsilon$ makes $\rho(n,m) < \epsilon$ for all $n, m \geq N$, so $(n)_{n\geq 1}$ is ρ-Cauchy. It has no ρ-limit: the condition $\rho(n,x) \to 0$ says $f(n) \to f(x)$ in $\mathbb{R}$, and $f(n) = n/(1+n) \to 1$, so proposition 1.2.6 would give $f(x) = 1$, which no real x satisfies since $|f(x)| = |x|/(1+|x|) < 1$.

The same sequence shows ρ and d are not strongly equivalent, and shows it without any computation of constants: strongly equivalent metrics have the same Cauchy sequences, because $c_1 d \leq \rho \leq c_2 d$ converts each Cauchy estimate into the other with ϵ rescaled, and $(n)_{n\geq 1}$ is ρ-Cauchy but not d-Cauchy.

A property that every homeomorphism preserves is what the word "topological" means, and completeness is not one. The mechanism is theorem 1.3.2(iii): continuity is expressible in open sets alone, so a homeomorphism carries the open sets of one space bijectively onto those of the other, and every notion built from open sets alone (closure, interior, boundary, density, convergence, continuity of maps into and out of the space) transfers along with them. Completeness does not transfer, so completeness is not built from open sets alone. It is a property of the metric, and example 2.1.9 shows the metric can be exchanged for another with the same open sets and the opposite verdict.

The consequence constrains every later completeness proof. Open sets, closures, preimages and continuity may still appear in one, but only alongside a hypothesis carrying metric information: theorem 2.1.8(i) reasons through closedness, and it also assumes the ambient space complete, which is not data the open sets record. Each completeness proof in this book therefore either estimates distances directly, producing an N from an ϵ, or imports the estimate from a space already known to be complete. What does transfer is isometry, which preserves every quantity computed from distances (definition 1.3.11) and so preserves Cauchy sequences, convergent sequences and completeness alike. The gap example 1.3.12 opened between the two notions of isomorphism is exactly what example 2.1.9 measures.

Exercise 2.1.1

1. A uniformly continuous map carries Cauchy sequences to Cauchy sequences: the δ that definition 1.3.4 gives for a given ϵ converts a Cauchy estimate for δ into one for ϵ. Show that such a map need not carry completeness to completeness: let $\mathbb{N} = \{1, 2, 3, \dots\} \subseteq \mathbb{R}$ and $Y = \{1/n \mid n \geq 1\} \subseteq \mathbb{R}$ carry the metrics inherited from $\mathbb{R}$, and consider $g\colon \mathbb{N} \to Y$ with $g(n) = 1/n$. Show that $\mathbb{N}$ is complete, that g is a Lipschitz bijection, and that Y is not complete. Then explain why no Cauchy sequence of $\mathbb{N}$ witnesses the failure.

2. Let $(x_n)_{n\geq 1}$ be a sequence in a metric space with $\sum_{n=1}^{\infty} d(x_n, x_{n+1}) < \infty$. Show that (x_n) is Cauchy. Show that the weaker hypothesis $d(x_n, x_{n+1}) \to 0$ does not suffice, using the partial sums $x_n = \sum_{k=1}^{n} 1/k$ in $\mathbb{R}$.

3. Let X carry the discrete metric of example 1.1.5. Show that a sequence in X is Cauchy if and only if it is eventually constant, and deduce that X is complete. Show also that every subset of X is closed, and check that this agrees with theorem 2.1.8(ii).

4. Let $d(x, y) = |x - y|$ and $d'(x, y) = \min(|x - y|, 1)$ on $\mathbb{R}$, the pair of example 1.4.8, which is topologically but not strongly equivalent. Show that d and d' nevertheless have exactly the same Cauchy sequences, and that $(\mathbb{R}, d')$ is complete. Conclude that strong equivalence is sufficient but not necessary for two metrics to share their Cauchy sequences.

5. Show that the Cantor set C of example 1.2.13 is complete. Verify that $t_n = \sum_{j=1}^{n} 2 \cdot 3^{1-2j}$ lies in C for every n, compute $\lim_n t_n$, and check directly that the limit lies in C. Then explain why $\mathbb{Q}$, which like C has empty interior in $\mathbb{R}$, fails to be complete.

6. Show that a metric space X is complete if and only if every Cauchy sequence in X has a convergent subsequence. Deduce that a metric space in which *every* sequence has a convergent subsequence is complete.

2.2 The Complete Spaces of Analysis

Section 2.1 settled what completeness says; which spaces satisfy it is a separate question, and one that has to be answered space by space. This section answers it for the spaces the rest of the book runs on: $\mathbb{R}^n$, the bounded functions on a set and the bounded continuous functions on a metric space under the sup metric, and the sequence spaces ℓ^p. It closes with a space that fails, and with the reading of that failure: one set of functions, complete under one metric and not under another.

The first case costs almost nothing, because the completeness of $\mathbb{R}$ is available from *Elementary Analysis* [1] and a point of $\mathbb{R}^n$ is a finite list of reals. What has to be checked is that the reduction to coordinates works for Cauchy sequences and not just for convergent ones, and section 1.4 gave both halves of that reduction.

Theorem 2.2.1: Completeness of Finite Products and of $\mathbb{R}^n$

Let (X, d_X) and (Y, d_Y) be complete metric spaces. Then $X \times Y$ is complete under each of the three product metrics of definition 1.4.3. Consequently $\mathbb{R}^n$ is complete under each of d_1, d_2 and d_∞.

***Proof*:**
By proposition 1.4.4 each of the three product metrics is bounded above and below by a constant multiple of $d_{\max}$, so rescaling the tolerance ϵ by that constant converts a Cauchy condition for one of them into a Cauchy condition for another; and by proposition 1.4.9 the three have the same convergent sequences. It is therefore enough to argue for $d_{\max}$.

Let $(m_k)_{k\geq 1}$ be Cauchy in $X \times Y$ for $d_{\max}$, and write $m_k = (x_k, y_k)$. Since

$$d_X(x_k, x_l) \leq d_{\max}(m_k, m_l) \qquad \text{and} \qquad d_Y(y_k, y_l) \leq d_{\max}(m_k, m_l)$$

every N witnessing the Cauchy condition for (m_k) witnesses it for (x_k) in X and for (y_k) in Y as well. Completeness of the factors gives limits $x_k \to x$ and $y_k \to y$, and proposition 1.4.9 converts the pair of coordinate convergences back into $m_k \to (x, y)$ in $X \times Y$.

For $\mathbb{R}^n$, induct on n. The case $n = 1$ is the completeness of $\mathbb{R}$ [1]. For the inductive step, identify $\mathbb{R}^n$ with $\mathbb{R}^{n-1} \times \mathbb{R}$. Under that identification the metric d_∞ of example 1.1.3 on $\mathbb{R}^n$ is exactly $d_{\max}$ built from d_∞ on $\mathbb{R}^{n-1}$ and $|\cdot|$ on $\mathbb{R}$, since a maximum over n numbers is the maximum of the first $n-1$ and the last. The first part of the theorem and the inductive hypothesis give completeness for d_∞, and the comparison $d_\infty \le d_2 \le d_1 \le n\,d_\infty$ recorded in corollary 1.4.7 transfers it to d_1 and d_2 by the same rescaling, completing the proof.

No new analysis entered that argument: the completeness of $\mathbb{R}^n$ is the completeness of $\mathbb{R}$ used n times. Two conveniences made it cheap, and both are consequences of the number of coordinates being finite. The candidate limit was assembled from finitely many separate limits, so no care was needed about uniformity in the coordinate; and once assembled it was automatically a point of the space, since every n-tuple of reals lies in $\mathbb{R}^n$. Neither convenience persists in the passage to infinitely many coordinates, and the proofs below are longer precisely where they fail.

Combining the theorem with theorem 2.1.8 produces many complete spaces at once: every closed subset of $\mathbb{R}^n$ is complete in the subspace metric. The closed unit ball, any sphere $S_r(x)$, and the Cantor set C of example 1.2.13, which is closed as an intersection of closed sets, are all complete metric spaces in their own right. The rationals $\mathbb{Q}^n$ are not, being dense in $\mathbb{R}^n$ without being closed.

$\mathbb{R}^n$ was cheap because a point of it is a finite list. The spaces that follow have points that are functions, and there the data attached to a single point is already infinite. The largest such space requires nothing of its functions but boundedness, and it is proved complete first because the two spaces that matter both sit inside it: the bounded continuous functions as a closed subspace, and ℓ^∞ as an equality.

For a nonempty set X, let $B(X)$ denote the set of bounded functions $f\colon X \to \mathbb{R}$ and put

$$d_\infty(f,g) = \sup_{x\in X} \bigl|f(x) - g(x)\bigr|$$

The supremum is finite because $f-g$ is bounded, and the verification that d_∞ is a metric is the argument of example 1.1.4 with $[a,b]$ replaced by X: that argument used only boundedness of the difference, never continuity or any structure on the domain. When X is a nonempty metric space, $C_b(X) \subseteq B(X)$ denotes the set of bounded continuous functions $X \to \mathbb{R}$, carrying the subspace metric of definition 1.4.1, which is again d_∞.

Theorem 2.2.2: Completeness of the Bounded Functions

Let X be a nonempty set. Then $\bigl(B(X), d_\infty\bigr)$ is complete.

***Proof*:**
Let $(f_n)_{n\ge1}$ be Cauchy in $B(X)$.

Step 1: manufacturing the candidate limit. Fix $x \in X$. Since $|f_n(x) - f_m(x)|$ is one of the numbers whose supremum is $d_\infty(f_n, f_m)$,

$$\bigl|f_n(x) - f_m(x)\bigr| \le d_\infty(f_n, f_m)$$

so the real sequence $\bigl(f_n(x)\bigr)_{n\ge1}$ is Cauchy, and by the completeness of $\mathbb{R}$ [1] it converges. Define

$$f(x) = \lim_{n\to\infty} f_n(x)$$

which specifies a function $f\colon X \to \mathbb{R}$. It is the only possible limit: if $g \in B(X)$ satisfied $d_\infty(f_n, g) \to 0$ then the displayed inequality with g in place of f_m would force $f_n(x) \to g(x)$ at every x, and limits of real sequences are unique.

Step 2: the candidate lies in the space. Apply the Cauchy condition with $\epsilon = 1$ to get an N with $d_\infty(f_n, f_N) \le 1$ for all $n \ge N$. Writing $M = \sup_x |f_N(x)|$, which is finite because f_N is bounded, every $x \in X$ and every $n \ge N$ satisfy

$$\big|f_n(x)\big| \le \big|f_n(x) - f_N(x)\big| + \big|f_N(x)\big| \le 1 + M$$

Letting $n \to \infty$ at the fixed point x preserves the non-strict inequality, so $|f(x)| \le 1 + M$ for every $x \in X$. Hence f is bounded and $f \in B(X)$.

Step 3: convergence in the metric. This is the step where the order in which the quantifiers are chosen decides the argument. Let $\epsilon > 0$ and choose N so that

$$d_\infty(f_n, f_m) \le \frac{\epsilon}{2} \qquad \text{for all } n, m \ge N$$

This N is fixed now, before any point of X is named. Fix $n \ge N$ and $x \in X$. For every $m \ge N$,

$$\big|f_n(x) - f_m(x)\big| \le d_\infty(f_n, f_m) \le \frac{\epsilon}{2}$$

As $m \to \infty$ the left-hand side converges to $|f_n(x) - f(x)|$, by Step 1 and the continuity of the metric on $\mathbb{R}$ (proposition 1.3.8), and a non-strict inequality survives that limit:

$$\big|f_n(x) - f(x)\big| \le \frac{\epsilon}{2}$$

The point x was arbitrary and N did not depend on it, so this bound holds at every point of X simultaneously and $\epsilon/2$ is an upper bound for the set whose supremum is $d_\infty(f_n, f)$. Therefore $d_\infty(f_n, f) \le \epsilon/2 < \epsilon$ for all $n \ge N$, which says $f_n \to f$ in $B(X)$, completing the proof.

The three steps are the template for every completeness proof about a space of functions or sequences, and the later ones in this section follow it without deviation. Step 1 uses nothing about the space beyond the completeness of $\mathbb{R}$: the limit is built one point at a time, and at that stage it is only a function, with no claim on membership. Step 2 checks the defining property of the space, here boundedness, and it is the step that fails when the space is chosen badly. Step 3 upgrades convergence at each point separately to convergence in the metric, and the mechanism of the upgrade is that N is chosen from the Cauchy condition before x is named. Choosing N after x would give a bound at each point and no bound on the supremum, which is exactly the difference between a sequence of functions converging at every point and converging in d_∞. **??** makes that difference its subject.

The space of bounded real sequences is an instance rather than an analogue. A sequence is a function $\mathbb{N} \to \mathbb{R}$; it is bounded as a sequence exactly when it is bounded as a function; and the metric $d_\infty(x, y) = \sup_k |x_k - y_k|$ of example 1.1.9 is the sup metric of $B(\mathbb{N})$ written in sequence notation. So $\ell^\infty = B(\mathbb{N})$ as metric spaces, and theorem 2.2.2 already proves ℓ^∞ complete.

The set $B(X)$ is large and has no analytic content of its own, since it requires nothing of its functions except that their values not run away. The space analysis works in is the continuous part of it, and the passage from one to the other is handled by theorem 2.1.8: rather than running the three steps again, it is enough to show that $C_b(X)$ is closed in $B(X)$. That closedness is a statement with a name of its own, namely that a limit in the sup metric of continuous functions is continuous, and the proof is the standard splitting of one estimate into three.

Theorem 2.2.3: Completeness of the Bounded Continuous Functions

Let X be a nonempty metric space. Then $C_b(X)$ is a closed subset of $\big(B(X), d_\infty\big)$, and $\big(C_b(X), d_\infty\big)$ is complete. In particular $C[a,b]$ with the metric of example 1.1.4 is complete.

***Proof*:**
By corollary 1.2.10 it suffices to show that whenever $f_n \in C_b(X)$ and $f_n \to f$ in $B(X)$, the limit f is continuous, since f is then bounded and continuous and so lies in $C_b(X)$.

Fix $x_0 \in X$ and $\epsilon > 0$. Choose n with $d_\infty(f_n, f) < \epsilon/3$, which is possible because $d_\infty(f_n, f) \to 0$, and hold that n fixed. Continuity of f_n at x_0 gives a $\delta > 0$ such that $d(x, x_0) < \delta$ implies $|f_n(x) - f_n(x_0)| < \epsilon/3$. For any such x, inserting f_n twice and applying the triangle inequality in $\mathbb{R}$ gives

$$\big|f(x) - f(x_0)\big| \leq \big|f(x) - f_n(x)\big| + \big|f_n(x) - f_n(x_0)\big| + \big|f_n(x_0) - f(x_0)\big|$$

The first and third terms are each at most $d_\infty(f, f_n) < \epsilon/3$, and the middle term is less than $\epsilon/3$ by the choice of δ, so the sum is less than ϵ. Hence f is continuous at x_0, and since x_0 was arbitrary, $f \in C_b(X)$. So $C_b(X)$ is closed in $B(X)$, and theorem 2.2.2 together with theorem 2.1.8 makes it complete.

For the last claim, take $X = [a,b]$. Every continuous function on a closed bounded interval is bounded [1], so $C[a,b] = C_b([a,b])$ as sets, and the metric of example 1.1.4 is d_∞, completing the proof.

The two outer terms in the displayed estimate are controlled at the point x and at the point x_0 by the same number $d_\infty(f, f_n)$, and that is the whole content of the argument. A bound valid at each point separately would not do: the first term is evaluated at a point x that is only produced after n has been chosen, so any control over it has to be uniform in the point. This is why the theorem is false for pointwise limits. On $[0,1]$ the functions $f_n(x) = x^n$ are continuous and converge at every point, to 0 for $x < 1$ and to 1 at $x = 1$, and the limit is not continuous. Consistently with theorem 2.2.3, the sequence (x^n) is not Cauchy in $C[0,1]$: for any n, substituting $t = x^n$, which ranges over all of $[0,1]$ as x does,

$$d_\infty(f_n, f_{2n}) = \sup_{x \in [0,1]} \big|x^n - x^{2n}\big| = \max_{t \in [0,1]} (t - t^2) = \frac{1}{4}$$

attained at $t = \frac{1}{2}$, that is, at $x = 2^{-1/n}$. A Cauchy sequence in $C[0,1]$ cannot have two terms at distance $\frac{1}{4}$ arbitrarily far out.

Also complete, by the same theorem, is every closed subset of $C[a,b]$: the functions vanishing at a prescribed point, the functions with values in $[0,1]$, the functions with a prescribed integral. Each is complete because it is closed, and each is closed because it is cut out by conditions preserved under convergence in d_∞.

The sequence spaces ℓ^p with p finite are the remaining running example, and they are where the two conveniences of theorem 2.2.1 both fail. The coordinatewise argument still produces a candidate limit, but that candidate need not lie in the space: the points $x^{(k)} = (1, \ldots, 1, 0, 0, \ldots)$ with k ones all lie in ℓ^1 and converge in each coordinate to $(1, 1, 1, \ldots)$, which lies in no ℓ^p with p finite. What rules such behaviour out for a *Cauchy* sequence is a bound on the tails that is uniform in k, and producing that bound is the content of the next proof.

Theorem 2.2.4: Completeness of the Sequence Spaces

Let $1 \leq p < \infty$. Then (ℓ^p, d_p) is complete, and (ℓ^∞, d_∞) is complete.

***Proof*:**
The case of ℓ^∞ is theorem 2.2.2 in different notation: $\ell^\infty = B(\mathbb{N})$ as metric spaces, since a bounded sequence is a bounded function $\mathbb{N} \to \mathbb{R}$ and the two metrics are the same supremum.

Fix $1 \leq p < \infty$ for the remainder, and let $\big(x^{(k)}\big)_{k\geq 1}$ be Cauchy in ℓ^p, with $x^{(k)} = \big(x_1^{(k)}, x_2^{(k)}, \dots\big)$.

Step 1: the coordinatewise limit. Fix an index j. A single term of a sum of nonnegative terms is at most the whole sum, so

$$\big|x_j^{(k)} - x_j^{(l)}\big| = \Big(\big|x_j^{(k)} - x_j^{(l)}\big|^p\Big)^{1/p} \leq \left(\sum_{i=1}^{\infty} \big|x_i^{(k)} - x_i^{(l)}\big|^p\right)^{1/p} = d_p\big(x^{(k)}, x^{(l)}\big)$$

Hence $\big(x_j^{(k)}\big)_{k\geq 1}$ is Cauchy in $\mathbb{R}$ and converges; set $x_j = \lim_k x_j^{(k)}$ and $x = (x_j)_{j\geq 1}$.

Step 2: the tail estimate, and membership in the space. Let $\epsilon > 0$ and choose N with $d_p\big(x^{(k)}, x^{(l)}\big) \leq \epsilon$ for all $k, l \geq N$. Fix $k \geq N$ and a finite $J \geq 1$. For every $l \geq N$, discarding the terms beyond J from a sum of nonnegative terms gives

$$\sum_{j=1}^{J} \big|x_j^{(k)} - x_j^{(l)}\big|^p \leq d_p\big(x^{(k)}, x^{(l)}\big)^p \leq \epsilon^p \tag{2.1}$$

The left-hand side of (2.1) is a sum of finitely many terms, each of which converges as $l \to \infty$ by Step 1, so letting $l \to \infty$ preserves the bound:

$$\sum_{j=1}^{J} \big|x_j^{(k)} - x_j\big|^p \leq \epsilon^p$$

This holds for every J, and the left-hand side is nondecreasing in J, so the series converges and its sum obeys the same bound:

$$\sum_{j=1}^{\infty} \big|x_j^{(k)} - x_j\big|^p \leq \epsilon^p \qquad \text{for all } k \geq N \tag{2.2}$$

In particular the sequence $x^{(k)} - x$ lies in ℓ^p. Since $x^{(k)} \in \ell^p$ and ℓ^p is closed under differences by theorem 1.1.8, the identity $x = x^{(k)} - \big(x^{(k)} - x\big)$ places x in ℓ^p.

Step 3: convergence in the metric. Taking pth roots in (2.2) reads $d_p\big(x^{(k)}, x\big) \leq \epsilon$ for all $k \geq N$. Since $\epsilon > 0$ was arbitrary and N was chosen from it alone, $x^{(k)} \to x$ in ℓ^p, completing the proof.

The proof is theorem 2.2.2 again with "bounded" replaced by "p-summable": a limit built one coordinate at a time, a check that it lies in the space, and an upgrade from coordinatewise convergence to convergence in the metric. The device carrying Step 2 is worth isolating, since it recurs whenever a statement about an infinite sum has to be extracted from statements about its terms. A bound on the full sum is inherited by each finite truncation; a finite truncation involves finitely many limits and so passes to the limit termwise; and the resulting bound, being independent of where the truncation was made, passes back to the full sum by monotone convergence of the partial sums. No interchange of two infinite limits is ever performed, which is what makes the argument elementary.

Steps 2 and 3 also came out of a single computation, (2.2), where in theorem 2.2.2 they were separate. That is a feature of the norm rather than a shortcut: the quantity that has to be small for convergence, and the quantity that has to be finite for membership, are the same sum.

Every space in this section so far has been complete. The next one is not, and it differs from the incomplete spaces of example 2.1.7 in the respect that matters here: its underlying set is $C[0,1]$, exactly the set that theorem 2.2.3 has just proved complete, and the only thing changed is the metric.

Example 2.2.5: Incompleteness of $C[0,1]$ in the L^1 Metric

On $C[0,1]$ define

$$d_1(f,g) = \int_0^1 \big|f(x) - g(x)\big|\,dx$$

The subscript repeats the one carried by the ℓ^1 metric of example 1.1.9 and by the taxicab metric on $\mathbb{R}^n$, the pattern in each case being a sum or an integral of first powers; the ambient space says which is meant. The integral exists because $|f-g|$ is continuous, hence Riemann integrable [1]. Symmetry is clear, the triangle inequality follows from $|f-h| \le |f-g| + |g-h|$ pointwise and monotonicity of the integral, and the separation axiom is the following observation, used twice more below: if h is continuous on an interval $[a,b]$ with $a<b$ and $\int_a^b |h| = 0$, then h vanishes identically there. Otherwise $|h(x_0)| = c > 0$ for some $x_0 \in [a,b]$, and continuity gives an $\eta > 0$ with $|h| > c/2$ on an interval $J \subseteq [a,b]$ of length at least η containing x_0, whence $\int_a^b |h| \ge \int_J |h| \ge c\eta/2 > 0$.

For $n \ge 2$ let $f_n\colon [0,1] \to \mathbb{R}$ be the ramp of width $1/n$ that climbs from 0 to 1 just to the right of $\frac{1}{2}$:

$$f_n(x) = \begin{cases} 0 & 0 \le x \le \frac{1}{2} \\ n\big(x - \frac{1}{2}\big) & \frac{1}{2} \le x \le \frac{1}{2} + \frac{1}{n} \\ 1 & \frac{1}{2} + \frac{1}{n} \le x \le 1 \end{cases}$$

The three formulas agree at the two joins, giving 0 at $x = \frac{1}{2}$ and 1 at $x = \frac{1}{2} + \frac{1}{n}$, so each f_n is continuous.

The sequence is Cauchy. For $m \ge n \ge 2$ the functions f_n and f_m agree on $[0, \frac{1}{2}]$, where both vanish, and on $[\frac{1}{2} + \frac{1}{n}, 1]$, where both equal 1. On the remaining interval, of length $1/n$, both take values in $[0,1]$, so their difference is at most 1 in absolute value. Hence

$$d_1(f_n, f_m) = \int_{1/2}^{1/2+1/n} \big|f_n - f_m\big| \le \frac{1}{n}$$

and given $\epsilon > 0$ any $N > 1/\epsilon$ witnesses the Cauchy condition.

No continuous function is its limit. Suppose $f \in C[0,1]$ satisfied $d_1(f_n, f) \to 0$. Since f_n vanishes on $[0, \frac{1}{2}]$,

$$\int_0^{1/2} |f| = \int_0^{1/2} \big|f - f_n\big| \le d_1(f_n, f)$$

for every n; the left-hand side does not depend on n, so it is 0, and by the observation above f vanishes identically on $[0, \frac{1}{2}]$. Now fix $\eta \in (0, \frac{1}{2})$. For every $n > 1/\eta$ the function f_n equals 1 on

$[\frac{1}{2}+\eta, 1]$, so the same argument gives

$$\int_{1/2+\eta}^{1} |f-1| = \int_{1/2+\eta}^{1} |f-f_n| \le d_1(f_n, f)$$

and hence $f \equiv 1$ on $[\frac{1}{2}+\eta, 1]$. As η was arbitrary in $(0, \frac{1}{2})$, $f \equiv 1$ on $(\frac{1}{2}, 1]$. But then $f(\frac{1}{2}) = 0$ while $f(\frac{1}{2}+\frac{1}{k}) = 1$ for all large k, so f does not preserve the convergence $\frac{1}{2}+\frac{1}{k} \to \frac{1}{2}$ and is not continuous at $\frac{1}{2}$ by theorem 1.3.2(ii). This contradiction shows that $\big(C[0,1], d_1\big)$ is not complete.

The sequence is not failing to converge for want of trying. Let g be the step function equal to 0 on $[0, \frac{1}{2}]$ and to 1 on $(\frac{1}{2}, 1]$, which is bounded with a single discontinuity and so is Riemann integrable. The same computation as above gives $\int_0^1 |f_n - g| \le 1/n$, so in any space of integrable functions containing g the ramps do converge, and they converge to g. The limit exists; it is the space that is too small to hold it.

Two conclusions follow. The first is about completeness itself. The set $C[0,1]$ is complete under d_∞ by theorem 2.2.3 and incomplete under d_1 by example 2.2.5, with not a single function added or removed. Completeness is therefore a property of the pair (X, d) and not of the underlying set: nothing about a set of points, on its own, determines whether its Cauchy sequences converge. That the topology does not determine it either is the separate content of example 2.1.9, where the two metrics on $\mathbb{R}$ induce the same open sets and still disagree about completeness. The two metrics compared here do not induce the same open sets, so the present example does not reach that far; what it adds is that the phenomenon already occurs on the space analysis works in, between the two metrics that space is most often given.

The second is about what to do with a space that fails. There are two repairs. One is abstract and applies to every metric space at once: adjoin the missing limits formally, which is the completion constructed in section 2.4. The other is to identify the completion concretely, and for d_1 that means enlarging $C[0,1]$ to a space of functions closed under the limits the metric requires. The enlargement cannot be carried out with the Riemann integral, whose class of integrable functions is not closed under such limits either; building an integral whose function space is complete is the business of measure theory, and the space that results is called $L^1[0,1]$.[1] The failure exhibited here is a large part of why that theory exists.

Exercise 2.2.1

1. Show that a subset $A \subseteq \mathbb{R}^n$ is complete in the subspace metric if and only if it is closed in $\mathbb{R}^n$. Which of $\mathbb{Q} \cap [0,1]$, the Cantor set, and $\big\{x \in \mathbb{R}^2 \bigm| \|x\| < 1\big\}$ are complete?

2. For $n \ge 0$ let $f_n(x) = \sum_{k=0}^{n} \frac{x^k}{k!}$, a polynomial and hence an element of $C[0,1]$. Compute $d_\infty(f_n, f_m)$ for $m > n$ exactly, show that (f_n) is Cauchy in $\big(C[0,1], d_\infty\big)$, and identify its limit.

3. Let X be a nonempty set and $a \in X$. Show that $\{f \in B(X) \mid f(a) = 0\}$ is complete under d_∞. Show that $\{f \in B(X) \mid \sup_{x\in X} |f(x)| < 1\}$ is not.

4. In ℓ^1, exhibit a sequence of points converging in every coordinate to a point of ℓ^1 but not converging in d_1, and a sequence of points converging in every coordinate to a sequence lying in no ℓ^p with p finite. Verify directly that neither sequence is Cauchy in d_1, as theorem 2.2.4

[1] *Measure Theory* [3] constructs the Lebesgue integral and proves L^1 complete; that volume depends on this one, so the pointer runs forward and nothing below relies on it.

requires.

5. Regard ℓ^1 as a subset of ℓ^∞ and give it the metric d_∞. Show that (ℓ^1, d_∞) is not complete, using the points $y^{(k)} = \left(1, \frac{1}{2}, \ldots, \frac{1}{k}, 0, 0, \ldots\right)$. Reconcile this with theorem 2.2.4.

6. Show that $d_1(f,g) \leq d_\infty(f,g)$ for all $f, g \in C[0,1]$, where d_1 is the metric of example 2.2.5. Deduce from example 2.2.5 and theorem 2.2.3 that d_1 and d_∞ are not strongly equivalent. Then show they are not topologically equivalent either, by proving that $\{f \in C[0,1] \mid d_\infty(f,0) < 1\}$ contains no d_1-ball about the zero function.

2.3 Cantor's Intersection Theorem

Completeness has so far been something to verify: a property a space either has or lacks, checked case by case. This section spends it. The mechanism is a way of pinning down a point by a sequence of ever-tighter constraints, no one of which names it, and then letting completeness give the point the constraints describe. Cantor's intersection theorem is that mechanism in its bare form, and two consequences follow from it: a characterization of completeness in terms of nested sets, and the fact that a complete space with no isolated points is too large to be listed.

The constraints will be sets, and "ever tighter" has to be measured. In a bare metric space the only measurement available is the largest distance between two points of the set.

Definition 2.3.1: Diameter of a Set

Let (X, d) be a metric space and $A \subseteq X$. The *diameter* of A is

$$\operatorname{diam}(A) = \sup\{d(a,b) \mid a, b \in A\}$$

where $\operatorname{diam}(A) = \infty$ if the set of distances is unbounded, and $\operatorname{diam}(\emptyset) = 0$.

Two properties are immediate from the definition and are used constantly below. The first is monotonicity: if $A \subseteq B$ then the supremum defining $\operatorname{diam}(A)$ is taken over a subset of the set defining $\operatorname{diam}(B)$, so $\operatorname{diam}(A) \leq \operatorname{diam}(B)$. The second is that $\operatorname{diam}(A) < \infty$ says exactly that a nonempty A fits inside a single ball, the empty set being contained in every ball already. Indeed, if $\operatorname{diam}(A) = D < \infty$ and $a_0 \in A$ is any point, then every $a \in A$ satisfies $d(a, a_0) \leq D$, so $A \subseteq \overline{B}_D(a_0)$; conversely if $A \subseteq \overline{B}_r(x)$ then any $a, b \in A$ satisfy $d(a,b) \leq d(a,x) + d(x,b) \leq 2r$, so $\operatorname{diam}(A) \leq 2r$, and the same bound holds for $A \subseteq B_r(x)$ because $B_r(x) \subseteq \overline{B}_r(x)$. Taking A to be the closed ball itself records the fact that

$$\operatorname{diam}\left(\overline{B}_r(x)\right) \leq 2r$$

in every metric space. Diameter is thus a quantitative refinement of boundedness, and the theorem below needs the quantity, not just the qualitative statement.

Example 2.3.2: Diameters in the Standard Spaces

(i) In $\mathbb{R}$ with $d(x,y) = |x-y|$, the interval $[a,b]$ has diameter $b-a$, attained at the endpoints, and so does (a,b), where the supremum is not attained. The closed ball $\overline{B}_r(x) = [x-r, x+r]$ has diameter exactly $2r$, and so does the open ball $B_r(x) = (x-r, x+r)$, so the bound of $2r$ recorded above is sharp on the real line.

(ii) Under the discrete metric of example 1.1.5, $\operatorname{diam}(A) = 1$ for every A with at least two points and $\operatorname{diam}(A) = 0$ otherwise. An uncountable set and a two-point set have the same diameter: the quantity measures separation, not size.

(iii) In $C[0,1]$ with the sup metric of example 1.1.4, the set $A = \{f \in C[0,1] \mid \sup_x |f(x)| \leq 1\}$ has diameter 2. It is at most 2 because $d_\infty(f,g) \leq \sup_x |f(x)| + \sup_x |g(x)| \leq 2$, and it is at least 2 because the constant functions 1 and -1 both lie in A and are at distance 2.

(iv) In the Cantor set of example 1.2.13, each of the 2^n constituent intervals of C_n has diameter 3^{-n}, while $\operatorname{diam}(C) = 1$ because 0 and 1 both survive every stage. The diameters that tend to zero are the pieces, not the set.

Diameter interacts with closure in the simplest possible way, and the next lemma is what lets a hypothesis about closed sets be arranged rather than assumed.

Lemma 2.3.3: Diameter of a Closure

For every $A \subseteq X$, $\operatorname{diam}(\overline{A}) = \operatorname{diam}(A)$.

***Proof*:**
Since $A \subseteq \overline{A}$, monotonicity gives $\operatorname{diam}(A) \leq \operatorname{diam}(\overline{A})$, and if $\operatorname{diam}(A) = \infty$ the two are equal. So assume $\operatorname{diam}(A) < \infty$ and let $x, y \in \overline{A}$ and $\epsilon > 0$. By theorem 1.2.9(ii) the balls $B_{\epsilon/2}(x)$ and $B_{\epsilon/2}(y)$ each meet A, say in a and b respectively. The triangle inequality applied twice gives

$$d(x,y) \leq d(x,a) + d(a,b) + d(b,y) < \frac{\epsilon}{2} + \operatorname{diam}(A) + \frac{\epsilon}{2}$$

Since $\epsilon > 0$ was arbitrary and $d(x,y)$ does not depend on it, $d(x,y) \leq \operatorname{diam}(A)$. Taking the supremum over $x, y \in \overline{A}$ gives $\operatorname{diam}(\overline{A}) \leq \operatorname{diam}(A)$, as we sought to show.

Passing to the closure therefore changes nothing metrically: a nested family of sets with diameters tending to zero can always be replaced by the family of their closures, which is still nested (closure preserves inclusion) and still has diameters tending to zero. What the replacement does not preserve is the intersection, and that discrepancy is exactly the content of the closedness hypothesis in the theorem below.

The theorem itself answers a question that single-variable analysis answers for intervals. A nested sequence of closed intervals in $\mathbb{R}$ whose lengths shrink to zero has exactly one common point [1], and that point is often the only description available of the number being constructed: it is how a decimal expansion specifies a real number, one digit narrowing the interval at a time. Nothing in that argument uses intervals, or $\mathbb{R}$, beyond the fact that Cauchy sequences converge.

Theorem 2.3.4: Cantor's Intersection Theorem

Let (X, d) be a complete metric space and let

$$A_1 \supseteq A_2 \supseteq A_3 \supseteq \cdots$$

be nonempty closed subsets of X with $\operatorname{diam}(A_n) \to 0$ as $n \to \infty$. Then $\bigcap_{n \geq 1} A_n$ consists of exactly one point.

***Proof*:**
Step 1: a sequence attached to the nest. Each A_n is nonempty, so choose a point $x_n \in A_n$ for every $n \geq 1$.

Step 2: the sequence is Cauchy. Let $\epsilon > 0$ and choose N with $\operatorname{diam}(A_N) < \epsilon$, which is possible because the diameters tend to zero. If $m, n \geq N$ then $A_m \subseteq A_N$ and $A_n \subseteq A_N$ by nesting, so both x_m and x_n lie in A_N and therefore

$$d(x_m, x_n) \leq \operatorname{diam}(A_N) < \epsilon$$

which is the Cauchy condition of definition 2.1.1.

Step 3: the limit lies in every set. By completeness there is an $x \in X$ with $x_n \to x$. Fix n. For every $m \geq n$ the point x_m lies in $A_m \subseteq A_n$, so the tail $(x_m)_{m \geq n}$ is a sequence in A_n converging to x. Since A_n is closed, corollary 1.2.10 puts $x \in A_n$. As n was arbitrary, $x \in \bigcap_n A_n$.

Step 4: there is no second point. Suppose $x, y \in \bigcap_n A_n$. For every n both points lie in A_n, so $d(x, y) \leq \operatorname{diam}(A_n)$, and the right side tends to 0. Hence $d(x, y) = 0$ and $x = y$ by the first metric axiom of definition 1.1.1, completing the proof.

Every hypothesis is used at an identifiable place. Nonemptiness gives the sequence in Step 1; nesting and the shrinking diameters make it Cauchy in Step 2; completeness converts the Cauchy sequence into an actual point in Step 3; closedness keeps that point inside each A_n in Step 3; and the shrinking diameters return in Step 4 to force uniqueness. The theorem is worth reading as a manufacturing process rather than a statement about intersections: it takes a descending list of constraints and returns a point meeting all of them, and the point need not be given by any formula. **??** manufactures a point in exactly this spirit, drawing on completeness directly rather than on the theorem, and what it produces is a fixed point.

Since each hypothesis has a job, dropping any one of them should break the conclusion, and it does.

Example 2.3.5: Nested Closed Sets with Empty Intersection

(i) *Without completeness.* Work in $\mathbb{Q}$ with the metric inherited from $\mathbb{R}$ and set

$$A_n = \left\{ q \in \mathbb{Q} \;\middle|\; |q - \sqrt{2}| \leq \tfrac{1}{n} \right\}$$

Each A_n is nonempty, since $\mathbb{Q}$ is dense in $\mathbb{R}$ [1]; each is closed in $\mathbb{Q}$, being the intersection with $\mathbb{Q}$ of a closed subset of $\mathbb{R}$ (proposition 1.4.2 applied to complements); the family is nested; and $\operatorname{diam}(A_n) \leq 2/n \to 0$. But a rational lying in every A_n would satisfy $|q - \sqrt{2}| \leq 1/n$ for all n, hence $q = \sqrt{2}$, which is not rational. So $\bigcap_n A_n = \emptyset$.

(ii) *Without closedness.* In $\mathbb{R}$, which is complete by theorem 2.2.1, put $A_n = (0, \frac{1}{n})$. These are nonempty, nested, and $\operatorname{diam}(A_n) = 1/n \to 0$, but a point x of the intersection would satisfy $0 < x < 1/n$ for every n, which is impossible. The closures $\overline{A}_n = [0, \frac{1}{n}]$ satisfy every hypothesis, and by lemma 2.3.3 they have the same diameters; their intersection is $\{0\}$, a point that belonged to none of the original sets.

(iii) *Without shrinking diameters.* In $\mathbb{R}$ put $A_n = [n, \infty)$. These are nonempty, closed and nested, with empty intersection, and every one has infinite diameter.

(iv) *Without shrinking diameters, second version.* Finite diameter is not enough either. In ℓ^2 let e_k denote the sequence with a 1 in position k and zeros elsewhere, and put $A_n = \{e_k \mid k \geq n\}$. Any two distinct members are at distance $\sqrt{2}$, so a convergent sequence drawn from A_n is Cauchy and hence eventually constant, which puts its limit in A_n; by corollary 1.2.10 each A_n is closed. The family is nested and nonempty, and $\operatorname{diam}(A_n) = \sqrt{2}$ for every n. Yet $e_k \notin A_{k+1}$, so no point survives all stages and the intersection is empty.

Item (iii) shows that the diameter condition cannot simply be dropped, and item (iv) shows more, that it cannot even be relaxed to boundedness in a complete space: the sets of unit vectors in ℓ^2 are closed, nested, nonempty and of constant diameter $\sqrt{2}$, and they close in on nothing.

The theorem used completeness, and the failure in $\mathbb{Q}$ shows that completeness is not decoration there. In fact the nesting property is strong enough to recover completeness on its own, which turns theorem 2.3.4 from a consequence of completeness into an equivalent formulation of it.

Corollary 2.3.6: Completeness from Nested Sets

A metric space X is complete if and only if every nested sequence $A_1 \supseteq A_2 \supseteq \cdots$ of nonempty closed subsets of X with $\operatorname{diam}(A_n) \to 0$ has nonempty intersection.

***Proof*:**
If X is complete then theorem 2.3.4 gives the intersection property, with the stronger conclusion that the intersection is a single point.

For the converse, assume the intersection property and let $(x_n)_{n\geq 1}$ be a Cauchy sequence in X. Define the tails

$$T_n = \{x_m \mid m \geq n\} \qquad A_n = \overline{T}_n$$

Each A_n is closed and nonempty, and $T_{n+1} \subseteq T_n$ gives $A_{n+1} \subseteq A_n$, so the family is nested. For the diameters, let $\epsilon > 0$ and use the Cauchy condition to choose N with $d(x_m, x_k) < \epsilon/2$ for all $m, k \geq N$. Then $\operatorname{diam}(T_N) \leq \epsilon/2$, and lemma 2.3.3 gives $\operatorname{diam}(A_N) = \operatorname{diam}(T_N) \leq \epsilon/2 < \epsilon$; by monotonicity $\operatorname{diam}(A_n) \leq \operatorname{diam}(A_N) < \epsilon$ for every $n \geq N$. Hence $\operatorname{diam}(A_n) \to 0$.

The hypothesis now gives a point $x \in \bigcap_n A_n$, and this x is the limit of the sequence. Given $\epsilon > 0$, choose N with $\operatorname{diam}(A_N) < \epsilon$. For $n \geq N$ both $x_n \in T_N \subseteq A_N$ and $x \in A_N$, so $d(x_n, x) \leq \operatorname{diam}(A_N) < \epsilon$. Thus every Cauchy sequence converges and X is complete, completing the proof.

The converse is the direction that gets used. Verifying completeness of a space reduces to exhibiting, for each shrinking nest of closed sets, one point common to all of them, with no Cauchy sequence mentioned. The proof also explains the choice of sets: the closed tails of a Cauchy sequence are the canonical shrinking nest, and lemma 2.3.3 is what makes their diameters shrink at the same rate as the Cauchy estimate.

The second consequence of theorem 2.3.4 is a statement about size. A countable space can be listed, and a list is a weak kind of description; the theorem manufactures points that no list anticipates, provided the space has enough room to keep dodging. The condition that gives the room is that no point of the space is separated from the rest.

Definition 2.3.7: Isolated Points and Perfect Spaces

A point x of a metric space X is *isolated* if $B_r(x) = \{x\}$ for some $r > 0$. The space X is *perfect* if it is nonempty and has no isolated point.
A subset $A \subseteq X$ is *perfect* if it is closed in X and perfect as a subspace (definition 1.4.1).

A point x fails to be isolated exactly when every ball about it meets $X \setminus \{x\}$, which by definition 1.2.8 says that x is a limit point of X. So a space is perfect precisely when it is nonempty and $X = X'$, every one of its points being approachable from within the space. For a subset the two readings agree: by proposition 1.4.2 the ball $B_r^A(a)$ is $B_r(a) \cap A$, so a is a limit point of A in the subspace exactly when it is one in X, and a closed set A with $A \subseteq A'$ satisfies $A = A'$ by theorem 1.2.9.

Example 2.3.8: Spaces With and Without Isolated Points

(i) $\mathbb{R}$ is perfect, since for any x and any $r > 0$ the point $x + r/2$ lies in $B_r(x)$ and differs from x. So is any interval $[a, b]$ with $a < b$: the two points $x + \frac{1}{2}\min(r, b - x)$ and $x - \frac{1}{2}\min(r, x - a)$ both lie in $[a, b] \cap B_r(x)$, and at least one differs from x, since x cannot equal both endpoints.

(ii) $\mathbb{Q}$ is perfect. Every ball about a rational contains a second rational, since $\mathbb{Q}$ is dense in $\mathbb{R}$ [1]. Yet $\mathbb{Q}$ is countable, so perfectness alone cannot force a space to be large.

(iii) $\mathbb{Z}$ with the metric inherited from $\mathbb{R}$ has every point isolated, since $B_{1/2}(n) = \{n\}$, so it is not perfect. It is complete, being closed in $\mathbb{R}$ (theorem 2.1.8 with theorem 2.2.1), so completeness alone cannot force a space to be large either.

(iv) ℓ^2 is perfect: given x and $r > 0$, the point $x + \frac{r}{2}e_1$, where e_1 is the sequence with a 1 in the first position and zeros elsewhere, lies in $B_r(x)$ and differs from x.

The rational and integer examples show that the theorem below needs both of its hypotheses, and it needs them for different reasons: perfectness is what gives a second point at each stage of the construction, and completeness is what turns the construction into a point of the space.

The construction runs on closed balls, which are closed sets by proposition 1.2.3. The distinction between $\overline{B}_r(y)$ and $B_r(y)$ matters in the proof: the sets fed to theorem 2.3.4 must be closed, while the room needed to fit the next ball inside the current one comes from the open ball.

Theorem 2.3.9: Complete Perfect Spaces Are Uncountable

Every complete perfect metric space is uncountable.

***Proof*:**
Suppose, for contradiction, that X is complete, perfect and countable, and let $x_1, x_2, x_3, \ldots$ be a list containing every point of X, with repetitions allowed.

Step 1: X has at least two points. It is nonempty, so fix $p \in X$. Since p is not isolated, $B_1(p)$ contains a point other than p.

Step 2: a shrinking nest of closed balls that dodges the list. Points $y_n \in X$ and radii r_n with $0 < r_n \leq 2^{-n}$ are constructed recursively so that for every $n \geq 1$,

$$x_n \notin \overline{B}_{r_n}(y_n) \qquad \text{and} \qquad \overline{B}_{r_{n+1}}(y_{n+1}) \subseteq B_{r_n}(y_n) \tag{2.3}$$

For $n = 1$, use Step 1 to pick $y_1 \in X$ with $y_1 \neq x_1$, and set $r_1 = \min\left(\frac{1}{2},\ \frac{1}{2}d(y_1, x_1)\right)$, which is positive. Then $d(y_1, x_1) \geq 2r_1 > r_1$, so $x_1 \notin \overline{B}_{r_1}(y_1)$.

Suppose y_n and $r_n > 0$ have been chosen. The point y_n is not isolated, so $B_{r_n}(y_n)$ contains a point other than y_n and therefore contains at least two points; pick $y_{n+1} \in B_{r_n}(y_n)$ with $y_{n+1} \neq x_{n+1}$. Set

$$\delta = r_n - d(y_n, y_{n+1}) > 0 \qquad r_{n+1} = \min\left(2^{-(n+1)},\ \frac{\delta}{2},\ \frac{d(y_{n+1}, x_{n+1})}{2}\right)$$

All three quantities in the minimum are positive, so $r_{n+1} > 0$, and $r_{n+1} \leq 2^{-(n+1)}$. Since $d(y_{n+1}, x_{n+1}) \geq 2r_{n+1} > r_{n+1}$, the point x_{n+1} lies outside $\overline{B}_{r_{n+1}}(y_{n+1})$. For the inclusion, let $d(z, y_{n+1}) \leq r_{n+1}$. Then

$$d(z, y_n) \leq d(z, y_{n+1}) + d(y_{n+1}, y_n) \leq \frac{\delta}{2} + \left(r_n - \delta\right) < r_n$$

so $z \in B_{r_n}(y_n)$. Both conditions of (2.3) hold.

Step 3: applying the intersection theorem. Put $A_n = \overline{B}_{r_n}(y_n)$. Each is nonempty, containing y_n, and closed by proposition 1.2.3. The second condition of (2.3) gives $A_{n+1} \subseteq B_{r_n}(y_n) \subseteq A_n$, so the family is nested, and the bound recorded after definition 2.3.1 gives $\operatorname{diam}(A_n) \leq 2r_n \leq 2^{1-n} \to 0$. Since X is complete, theorem 2.3.4 produces a point z lying in every A_n.

Step 4: the contradiction. Fix n. Then $z \in A_n$ while $x_n \notin A_n$ by the first condition of (2.3), so $z \neq x_n$. This holds for every n, so z appears nowhere in a list that was supposed to contain every point of X. Hence no such list exists and X is uncountable, completing the proof.

The construction is a diagonal argument carried out with balls instead of digits. Each stage is handed one point of the proposed list and shrinks the current ball so as to exclude it, and perfectness is what makes the shrinking possible: a ball around an isolated point contains only that point, leaving nowhere to retreat to when the list names it. Completeness then does the work that no finite stage can do, converting the infinite sequence of retreats into an actual point. Both hypotheses are unavoidable, by the examples of $\mathbb{Q}$ and $\mathbb{Z}$ in example 2.3.8.

Applied to $\mathbb{R}$, the theorem gives uncountability without any mention of decimal expansions: $\mathbb{R}$ is complete by theorem 2.2.1 and perfect by example 2.3.8. The same two lines apply to ℓ^2, complete by theorem 2.2.4 and perfect by example 2.3.8. The case worth working out in full is a space containing no interval at all.

Example 2.3.10: The Cantor Set Is Complete and Perfect

Let $C = \bigcap_{n\geq 0} C_n$ be the Cantor set of example 1.2.13, with the metric inherited from $\mathbb{R}$.

It is complete. It is closed in $\mathbb{R}$, being an intersection of closed sets, and $\mathbb{R}$ is complete by theorem 2.2.1, so theorem 2.1.8 applies.

It is perfect. It is nonempty, and every endpoint of every constituent interval of every C_n belongs to C: deleting an open middle third removes no endpoint of the interval it is taken from, and the endpoints of a constituent interval of C_n are again endpoints of constituent intervals of C_{n+1}, so they survive all later stages. Now let $x \in C$ and $r > 0$, and choose n with $3^{-n} < r$. The point x lies in one of the constituent intervals I of C_n, and $\operatorname{diam}(I) = 3^{-n}$ as computed in example 2.3.2. The interval I has two distinct endpoints, both in C, so at least one of them, say e, differs from x; and $|x - e| \leq \operatorname{diam}(I) = 3^{-n} < r$. Hence $B_r(x)$ meets $C \setminus \{x\}$, and no point of C is isolated.

By theorem 2.3.9 the Cantor set is uncountable. Together with the fact that C has empty interior (example 1.2.13), this makes C an uncountable subset of $\mathbb{R}$ containing no interval of positive length.

Example 1.2.13 asserted the uncountability of C on the grounds that its points are the reals in $[0, 1]$ with a base-3 expansion using only the digits 0 and 2, and theorem 2.3.4 is what makes that description precise. Given a sequence $a = (a_n)_{n\geq 1}$ with each $a_n \in \{0, 2\}$, define intervals recursively by $I_0(a) = [0, 1]$ and letting $I_n(a)$ be the left third of $I_{n-1}(a)$ when $a_n = 0$ and the right third when $a_n = 2$. Each $I_n(a)$ is a constituent interval of C_n, the family is nested and closed with $\operatorname{diam}(I_n(a)) = 3^{-n} \to 0$, and $\mathbb{R}$ is complete, so the theorem returns a single point

$$\bigcap_{n\geq 0} I_n(a) = \{x(a)\} \qquad x(a) = \sum_{n\geq 1} a_n 3^{-n}$$

which lies in $\bigcap_n C_n = C$. The formula for $x(a)$ is read off from the left endpoint of $I_n(a)$, which is $\sum_{k\leq n} a_k 3^{-k}$. Distinct sequences give distinct points: if a and b first differ at index n, then $I_n(a)$ and $I_n(b)$ are different constituent intervals of C_n and hence disjoint. Every point of C arises this way, since $x \in C$ lies in C_n for each n, hence in exactly one constituent interval of C_n, and those intervals are nested and so record a single sequence a with $x = x(a)$. The assignment $a \mapsto x(a)$ is therefore a bijection from $\{0, 2\}^{\mathbb{N}}$ onto C, which is the base-3 description of example 1.2.13, and theorem 2.3.4 is what converts an infinite sequence of left-or-right choices into a real number.

Exercise 2.3.1

1. Let $A, B \subseteq X$ be nonempty sets with $A \cap B \neq \emptyset$. Show that $\operatorname{diam}(A \cup B) \leq \operatorname{diam}(A) + \operatorname{diam}(B)$, and give two subsets of $\mathbb{R}$ with $A \cap B = \emptyset$ for which the inequality fails.
2. The bound $\operatorname{diam}\left(\overline{B}_r(x)\right) \leq 2r$ holds in every metric space. Show that equality holds in $\mathbb{R}^n$ with the Euclidean metric of example 1.1.3 for every $r > 0$, and that the inequality is strict for the discrete metric of example 1.1.5 when $r = 1$.
3. Let X be complete and let $A_1 \supseteq A_2 \supseteq \cdots$ be nonempty closed sets with $\operatorname{diam}(A_n) \to 0$, so that $\bigcap_n A_n = \{x\}$ by theorem 2.3.4. Show that $\sup_{y\in A_n} d(y, x) \to 0$, and deduce that *every* choice of points $x_n \in A_n$ produces a sequence converging to x.
4. Use corollary 2.3.6 to give a second proof that a closed subspace F of a complete metric space X is complete (theorem 2.1.8). Begin by showing that a subset of F that is closed in the subspace F is closed in X.

5. Let $E \subseteq C$ be the set of all endpoints of all constituent intervals of all the C_n, $n \geq 0$. Show that E is countable and dense in C in the sense of definition 1.2.11, and deduce that C contains points that are not endpoints. Show that $1/4$ is one of them.

6. Show that every nonempty countable complete metric space has an isolated point. Exhibit such a space with exactly one point that is not isolated. Which hypothesis of theorem 2.3.9 fails for $\mathbb{Q}$?

2.4 The Completion

Incompleteness is a defect that can always be repaired. A space that is missing the limits of some of its Cauchy sequences can be enlarged by adjoining those limits, and the enlargement is not a matter of taste: it is determined, up to isometry, by the space it repairs. This section builds the enlargement out of the Cauchy sequences themselves and proves both halves of that claim.

The construction puts X inside a larger space, so the word for the placement cannot be "isometry" in the sense of definition 1.3.11, which requires a bijection. What is wanted is the distance-preserving condition with surjectivity dropped.

Definition 2.4.1: Isometric Embedding

Let (X, d_X) and (Y, d_Y) be metric spaces. A map $f\colon X \to Y$ is an *isometric embedding* if

$$d_Y\big(f(x), f(t)\big) = d_X(x,t) \qquad \text{for all } x, t \in X$$

No surjectivity is required.

An isometric embedding is injective for free: if $f(x) = f(t)$ then $d_X(x,t) = d_Y(f(x), f(t)) = 0$, and the first axiom of definition 1.1.1 forces $x = t$. It is also continuous, being Lipschitz with constant 1 (definition 1.3.5) and hence uniformly continuous by proposition 1.3.6. Giving the image $f(X)$ the subspace metric (definition 1.4.1), the corestriction $X \to f(X)$ is a bijection preserving distances, so an isometric embedding is exactly an isometry onto its image, and definition 1.3.11 is recovered as the case $f(X) = Y$.

Three instances are already at hand. The inclusion $\mathbb{Q} \hookrightarrow \mathbb{R}$ preserves $|x - y|$ by definition, so it is an isometric embedding with non-surjective image. The map $\mathbb{R} \to \ell^2$ sending x to the sequence $(x, 0, 0, \dots)$ is another, since

$$d_2\big((x,0,0,\dots),(t,0,0,\dots)\big) = \big(|x-t|^2\big)^{1/2} = |x-t|$$

and its image is the set of sequences vanishing after the first entry. The inclusion $C[0,1] \hookrightarrow B([0,1])$ of the continuous functions into all bounded functions is a third: both sides carry $d_\infty(f,g) = \sup_x |f(x) - g(x)|$, and the formula does not change when the ambient set grows. Injectivity alone is not enough for any of this: the map $\mathbb{R} \to \mathbb{R}$, $x \mapsto 2x$, is a continuous bijection onto its image and doubles every distance, so it is a homeomorphism but not an isometric embedding.

With the vocabulary in place the repair can be stated. The requirement on $\widehat{X}$ is that it be complete, that it contain an isometrically embedded copy of X, and that it contain nothing further than what the limits of X's Cauchy sequences require. That last requirement is density: every point of $\widehat{X}$ is approachable from the copy of X.

Theorem 2.4.2: Existence of the Completion

Let (X, d) be a metric space. There exist a complete metric space $(\widehat{X}, \widehat{d})$ and an isometric embedding $\iota\colon X \to \widehat{X}$ whose image $\iota(X)$ is dense in $\widehat{X}$. Such a pair $(\widehat{X}, \iota)$ is a *completion* of X.

***Proof*:**
Step 1: the raw material. Let $\mathcal{C}$ be the set of all Cauchy sequences in X (definition 2.1.1). Every constant sequence is Cauchy, so $\mathcal{C}$ is nonempty whenever X is; in the degenerate case $X = \emptyset$ both $\mathcal{C}$ and the space built from it below are empty, and the theorem holds vacuously. A Cauchy sequence is a specification of a location by successive approximation, and $\mathcal{C}$ collects every such specification, whether or not X contains a point answering to it.

Step 2: identifying specifications of the same location. Two Cauchy sequences should name the same point of the enlarged space exactly when they crowd together. For $a = (a_n)_{n\geq 1}$ and $b = (b_n)_{n\geq 1}$ in $\mathcal{C}$ write

$$a \sim b \quad \text{if and only if} \quad d(a_n, b_n) \to 0$$

This is an equivalence relation on $\mathcal{C}$. Reflexivity holds because $d(a_n, a_n) = 0$ for every n, and symmetry because d is symmetric. For transitivity, suppose $a \sim b$ and $b \sim c$; the triangle inequality gives

$$0 \leq d(a_n, c_n) \leq d(a_n, b_n) + d(b_n, c_n)$$

and the right-hand side is a sum of two null sequences, hence null, so $d(a_n, c_n) \to 0$ and $a \sim c$. Write $[a]$ for the equivalence class of a and set $\widehat{X} = \mathcal{C}/{\sim}$.

Step 3: the distance between two classes. The definition to be made is

$$\widehat{d}\big([a], [b]\big) = \lim_{n\to\infty} d(a_n, b_n)$$

and it carries two obligations: that the limit exists, and that it depends only on the two classes and not on the representatives chosen.

For existence, put $t_n = d(a_n, b_n)$ and apply proposition 1.3.8 to the four points a_n, b_n, a_m, b_m:

$$|t_n - t_m| = \big|d(a_n, b_n) - d(a_m, b_m)\big| \leq d(a_n, a_m) + d(b_n, b_m)$$

Given $\epsilon > 0$, choose N_1 with $d(a_n, a_m) < \epsilon/2$ for $n, m \geq N_1$ and N_2 with $d(b_n, b_m) < \epsilon/2$ for $n, m \geq N_2$; for $n, m \geq \max(N_1, N_2)$ the display gives $|t_n - t_m| < \epsilon$. So (t_n) is a Cauchy sequence of real numbers, and $\mathbb{R}$ is complete [1], so $\lim_n t_n$ exists.

For independence of representatives, suppose $a \sim a'$ and $b \sim b'$. The same inequality, now applied to a_n, b_n, a'_n, b'_n, gives

$$\big|d(a_n, b_n) - d(a'_n, b'_n)\big| \leq d(a_n, a'_n) + d(b_n, b'_n)$$

whose right-hand side tends to 0 by hypothesis. Two convergent real sequences whose difference is null have the same limit, so the two candidate values of $\widehat{d}([a], [b])$ agree.

Step 4: $\widehat{d}$ is a metric. Each $d(a_n, b_n)$ is nonnegative by lemma 1.1.2, so the limit is nonnegative; symmetry is inherited term by term; and the triangle inequality follows by taking limits in

$$d(a_n, c_n) \leq d(a_n, b_n) + d(b_n, c_n)$$

all three limits existing by Step 3. That leaves the first axiom of definition 1.1.1, and it is the reason Step 2 passed to classes: $\widehat{d}([a],[b]) = 0$ says $\lim_n d(a_n, b_n) = 0$, which is precisely the relation $a \sim b$, that is, $[a] = [b]$. On $\mathcal{C}$ itself the formula $\lim_n d(a_n, b_n)$ satisfies every axiom except this one, and two distinct Cauchy sequences converging to the same place sit at distance zero from each other; the quotient collapses exactly the pairs the formula cannot separate.

Step 5: the embedding. For $x \in X$ let $\iota(x)$ be the class of the constant sequence at x,

$$\iota(x) = \big[(x, x, x, \dots)\big]$$

which is defined because constant sequences are Cauchy. For $x, y \in X$ the sequence $d(x,y), d(x,y), \dots$ is constant, so

$$\widehat{d}\big(\iota(x), \iota(y)\big) = \lim_{n\to\infty} d(x,y) = d(x,y)$$

and ι is an isometric embedding in the sense of definition 2.4.1.

Step 6: the image is dense. Let $[a] \in \widehat{X}$ with $a = (a_n)_{n\geq 1}$ Cauchy, and let $\epsilon > 0$. Choose N with $d(a_n, a_m) < \epsilon/2$ for all $n, m \geq N$. The single point $a_N \in X$ then satisfies

$$\widehat{d}\big(\iota(a_N), [a]\big) = \lim_{m\to\infty} d(a_N, a_m) \leq \frac{\epsilon}{2} < \epsilon$$

since every term with $m \geq N$ is bounded by $\epsilon/2$ and a limit inherits such a bound. So every ball in $\widehat{X}$ meets $\iota(X)$, and theorem 1.2.9(ii) gives $\overline{\iota(X)} = \widehat{X}$, which is density in the sense of definition 1.2.11.

Step 7: $\widehat{X}$ is complete. Let $(\xi_k)_{k\geq 1}$ be a Cauchy sequence in $\widehat{X}$. The argument produces a candidate limit out of X and then shows the sequence reaches it.

By Step 6, for each k there is a point $y_k \in X$ with

$$\widehat{d}\big(\iota(y_k), \xi_k\big) < \frac{1}{k}$$

Unwinding that step makes the choice explicit: writing ξ_k as the class of a Cauchy sequence $a^{(k)} = (a_n^{(k)})_{n\geq 1}$, the point y_k may be taken to be $a_{N_k}^{(k)}$ for a suitable index N_k, so that $(y_k)_{k\geq 1}$ is a diagonal choice through the doubly indexed array $a_n^{(k)}$, one entry taken from each of the given sequences.

That diagonal is Cauchy in X. Since ι preserves distances, the triangle inequality applied twice gives

$$\begin{aligned} d(y_k, y_l) &= \widehat{d}\big(\iota(y_k), \iota(y_l)\big) \\ &\leq \widehat{d}\big(\iota(y_k), \xi_k\big) + \widehat{d}(\xi_k, \xi_l) + \widehat{d}\big(\xi_l, \iota(y_l)\big) \\ &< \frac{1}{k} + \widehat{d}(\xi_k, \xi_l) + \frac{1}{l} \end{aligned}$$

Given $\epsilon > 0$, choose K_1 with $\widehat{d}(\xi_k, \xi_l) < \epsilon/3$ for $k, l \geq K_1$, which is possible because (ξ_k) is Cauchy, and K_2 with $1/K_2 < \epsilon/3$. For $k, l \geq \max(K_1, K_2)$ the display is less than ϵ. Hence $y = (y_k)_{k\geq 1}$ lies in $\mathcal{C}$ and $\xi = [y]$ is a point of $\widehat{X}$.

It remains to show $\xi_k \to \xi$. Let $\epsilon > 0$ and choose K with both $1/K < \epsilon/3$ and $d(y_k, y_m) < \epsilon/3$ for all $k, m \geq K$, the latter being available because y is Cauchy. For $k \geq K$, the class ξ is represented by $(y_m)_{m\geq 1}$ and $\iota(y_k)$ by the constant sequence at y_k, so

$$\widehat{d}\big(\iota(y_k), \xi\big) = \lim_{m\to\infty} d(y_k, y_m) \leq \frac{\epsilon}{3}$$

every term with $m \geq K$ obeying that bound. Combining this with the choice of y_k,

$$\widehat{d}(\xi_k, \xi) \leq \widehat{d}\big(\xi_k, \iota(y_k)\big) + \widehat{d}\big(\iota(y_k), \xi\big) < \frac{1}{k} + \frac{\epsilon}{3} < \epsilon$$

for every $k \geq K$. So every Cauchy sequence in $\widehat{X}$ converges, and $\widehat{X}$ is complete in the sense of definition 2.1.6, completing the proof.

The theorem answers the question of existence without saying anything about what the new points are. A point of $\widehat{X}$ is a rule for approaching a location by successive approximation, together with the decision to regard two rules as the same when they approach each other; nothing beyond the metric d goes into the construction, so nothing beyond it can come out. When $X = \mathbb{Q}$ with $d(x, y) = |x - y|$ this is the classical construction of the real numbers from the rationals, and the reader who has seen that construction has seen the whole of the proof above with X specialized.

Density is what keeps $\widehat{X}$ from being larger than the repair requires. A complete space containing an isometric copy of X need not resemble a completion of X at all, since it may contain a great deal besides that copy, and exercise 2.4.1 (6) exhibits the failure. Requiring the copy to be dense forces every point of $\widehat{X}$ to be the limit of points of X. Density by itself says nothing about completeness: $\mathbb{Q}$ contains a dense isometric copy of $\mathbb{Q}$, namely itself, and is not complete (example 2.1.7(i)). What Step 7 adds is that the space produced by the construction, which gives a point for every Cauchy sequence of X and for nothing else, is already complete: the limits demanded by X's Cauchy sequences are all the limits there are to require.

Two different constructions could produce two different-looking complete spaces around X, and for $\mathbb{Q}$ they historically did: Dedekind's cuts and Cantor's Cauchy sequences, both published in 1872, build the real numbers out of different raw material. The next theorem says the difference is immaterial. Any two completions of the same space are isometric, and by an isometry compatible with the two copies of X inside them, which is what licenses the phrase "the completion".

Theorem 2.4.3: Uniqueness of the Completion

Let (X, d) be a metric space and let $(\widehat{X}_1, \iota_1)$ and $(\widehat{X}_2, \iota_2)$ be completions of X, with metrics $\widehat{d}_1$ and $\widehat{d}_2$. There is exactly one continuous map $\Phi\colon \widehat{X}_1 \to \widehat{X}_2$ satisfying $\Phi \circ \iota_1 = \iota_2$, and it is an isometry in the sense of definition 1.3.11.

***Proof*:**

Step 1: constructing Φ. Let $\xi \in \widehat{X}_1$. Since $\iota_1(X)$ is dense, theorem 1.2.9(iii) gives a sequence (x_n) in X with $\iota_1(x_n) \to \xi$. A convergent sequence is Cauchy (proposition 2.1.2), and ι_1 preserves distances, so

$$d(x_n, x_m) = \widehat{d}_1\big(\iota_1(x_n), \iota_1(x_m)\big)$$

shows (x_n) is Cauchy in X. Applying ι_2, which also preserves distances, makes $\big(\iota_2(x_n)\big)$ Cauchy in $\widehat{X}_2$, and $\widehat{X}_2$ is complete, so the sequence converges. Define

$$\Phi(\xi) = \lim_{n\to\infty} \iota_2(x_n)$$

Step 2: the value does not depend on the approximating sequence. Suppose (x_n) and (x'_n) both satisfy $\iota_1(x_n) \to \xi$ and $\iota_1(x'_n) \to \xi$. By the consequence recorded in proposition 1.3.8,

$\widehat{d}_1\big(\iota_1(x_n), \iota_1(x'_n)\big) \to \widehat{d}_1(\xi, \xi) = 0$, and both embeddings preserve distances, so

$$\widehat{d}_2\big(\iota_2(x_n), \iota_2(x'_n)\big) = d(x_n, x'_n) = \widehat{d}_1\big(\iota_1(x_n), \iota_1(x'_n)\big) \to 0$$

If $\iota_2(x_n) \to \zeta$ and $\iota_2(x'_n) \to \zeta'$ then proposition 1.3.8 again gives $\widehat{d}_2(\zeta, \zeta') = \lim_n \widehat{d}_2\big(\iota_2(x_n), \iota_2(x'_n)\big) = 0$, so $\zeta = \zeta'$ and Φ is well defined.

Step 3: Φ preserves distances. Let $\xi, \eta \in \widehat{X}_1$ with approximating sequences (x_n) and (y_n), so that $\iota_2(x_n) \to \Phi(\xi)$ and $\iota_2(y_n) \to \Phi(\eta)$. Applying proposition 1.3.8 in $\widehat{X}_2$ and then in $\widehat{X}_1$,

$$\begin{aligned}\widehat{d}_2\big(\Phi(\xi), \Phi(\eta)\big) &= \lim_{n\to\infty} \widehat{d}_2\big(\iota_2(x_n), \iota_2(y_n)\big) = \lim_{n\to\infty} d(x_n, y_n) \\ &= \lim_{n\to\infty} \widehat{d}_1\big(\iota_1(x_n), \iota_1(y_n)\big) = \widehat{d}_1(\xi, \eta)\end{aligned}$$

the two outer equalities using the convergences above and the two inner ones the fact that ι_1 and ι_2 preserve distances. In particular Φ is an isometric embedding, hence injective and continuous.

Step 4: Φ is surjective. Let $\zeta \in \widehat{X}_2$. Density of $\iota_2(X)$ gives (x_n) in X with $\iota_2(x_n) \to \zeta$, and the argument of Step 1, run with the roles of the two completions exchanged, shows (x_n) is Cauchy in X and therefore that $\big(\iota_1(x_n)\big)$ is Cauchy in the complete space $\widehat{X}_1$. Let ξ be its limit. Then (x_n) is an approximating sequence for ξ, so $\Phi(\xi) = \lim_n \iota_2(x_n) = \zeta$.

Step 5: compatibility and uniqueness. For $x \in X$ the constant sequence $x_n = x$ satisfies $\iota_1(x_n) \to \iota_1(x)$, so $\Phi\big(\iota_1(x)\big) = \lim_n \iota_2(x) = \iota_2(x)$, which is the identity $\Phi \circ \iota_1 = \iota_2$. Now let $\Psi\colon \widehat{X}_1 \to \widehat{X}_2$ be any continuous map with $\Psi \circ \iota_1 = \iota_2$, let $\xi \in \widehat{X}_1$, and choose (x_n) with $\iota_1(x_n) \to \xi$. By theorem 1.3.2(ii),

$$\Psi(\xi) = \lim_{n\to\infty} \Psi\big(\iota_1(x_n)\big) = \lim_{n\to\infty} \iota_2(x_n) = \Phi(\xi)$$

so $\Psi = \Phi$. With Steps 3 and 4, Φ is a distance-preserving bijection, as we sought to show.

The mechanism of the proof is worth separating from the statement. A continuous map is determined by its values on a dense set, because every point is a limit of points where the values are already known; that is Step 5. A map defined on a dense set can be extended to the whole space when its target is complete and it does not distort distances too badly, because the Cauchy sequences it is fed have limits waiting for them; that is Step 1. Neither half uses anything particular to completions, and exercise 2.4.1 (4) states the general form: a uniformly continuous map from a dense subset into a complete space extends uniquely to the whole space.

In practice the construction of theorem 2.4.2 is never run. To identify the completion of a given space it suffices to exhibit some complete space in which it sits isometrically and densely, and theorem 2.4.3 then says that space is the completion. The examples below all follow that pattern.

Example 2.4.4: Completions of $\mathbb{Q}$, $(0,1)$ and $C[0,1]$

(i) The completion of $\mathbb{Q}$ with the metric $|x-y|$ is $\mathbb{R}$. The inclusion preserves distances by definition, $\mathbb{R}$ is complete [1], and $\mathbb{Q}$ is dense in $\mathbb{R}$, which is the case $n = 1$ of example 1.2.12(i). So the inclusion is a completion, and theorem 2.4.3 makes every other completion of $\mathbb{Q}$ isometric to $\mathbb{R}$ by an isometry restricting to the identity on $\mathbb{Q}$. The construction of theorem 2.4.2 is not being used here to build $\mathbb{R}$: this book takes $\mathbb{R}$ and its completeness as given, and Step 3 of that proof appealed to the completeness of $\mathbb{R}$ in order to define $\widehat{d}$ at all.

(ii) The completion of $(0,1)$ is $[0,1]$. The inclusion $(0,1) \hookrightarrow [0,1]$ preserves distances; $[0,1]$ is closed in $\mathbb{R}$ and so is complete by theorem 2.1.8; and $(0,1)$ is dense in $[0,1]$, since $\frac{1}{n} \to 0$ and $1-\frac{1}{n} \to 1$ while every other point of $[0,1]$ already lies in $(0,1)$. The completion therefore adds exactly the two missing endpoints. The interval $(0,1)$ is homeomorphic to $\mathbb{R}$, by composing $t \mapsto 2t-1$ with the inverse of the map in example 1.3.12, and $\mathbb{R}$ needs no points added at all; how much a space is missing is a question about its metric, not about its topology (example 2.1.9).

(iii) The completion of $\mathbb{Q}$ with the p-adic metric d_p of example 1.1.11 is the space $\mathbb{Q}_p$ of p-adic numbers, and it is strictly larger than $\mathbb{Q}$. To see this, choose digits $c_k \in \{0,1,\ldots,p-1\}$ for $k \geq 0$ and put $s_n = \sum_{k=0}^{n} c_k p^k$. For $m > n$ the difference $s_m - s_n = \sum_{k=n+1}^{m} c_k p^k$ is divisible by p^{n+1}, so $d_p(s_n, s_m) \leq p^{-(n+1)}$ and (s_n) is Cauchy. If a second digit string (c'_k) first differs from (c_k) at index j, then for $n \geq j$

$$s_n - s'_n = p^j \left((c_j - c'_j) + \sum_{k=j+1}^{n} (c_k - c'_k) p^{k-j} \right)$$

and the bracket is congruent to $c_j - c'_j \not\equiv 0$ modulo p, so $v_p(s_n - s'_n) = j$ exactly and $d_p(s_n, s'_n) = p^{-j}$ for every $n \geq j$. The two Cauchy sequences are therefore inequivalent and determine distinct points of the completion. There are uncountably many digit strings and only countably many rationals, so $\mathbb{Q}_p$ is uncountable while the image of $\mathbb{Q}$ inside it is countable: all but countably many points of $\mathbb{Q}_p$ are new. The arithmetic of $\mathbb{Q}_p$, and the theory built on it, belong to *Local Field Theory* [2].

(iv) The completion of $C[0,1]$ under the metric $\int_0^1 |f-g|$ of example 2.2.5 is a space of integrable functions on $[0,1]$, whose points cannot be described as functions without a theory of integration more general than Riemann's, and are left undescribed here.[a]

[a]The theory in question is Lebesgue's: the completion is the space of Lebesgue integrable functions on $[0,1]$, with two functions counted as the same point when they differ only on a set of measure zero. Measure and the Lebesgue integral are not built in this book.

Exercise 2.4.1

1. Let X be complete and let $\iota\colon X \to \widehat{X}$ be as in theorem 2.4.2. Show that ι is surjective, so that a complete space is its own completion.

2. Let X be a complete metric space and let $A \subseteq X$ carry the subspace metric of definition 1.4.1. Show that the inclusion $A \hookrightarrow \overline{A}$ exhibits $\overline{A}$ as a completion of A. Deduce that the completion of $\mathbb{Q} \cap (0,1)$ is $[0,1]$.

3. Let $(\widehat{X}, \iota)$ be a completion of X. Show that $\operatorname{diam} \widehat{X} = \operatorname{diam} X$ for the diameter of definition 2.3.1, so that X is bounded if and only if $\widehat{X}$ is, with the same diameter.

4. Let $D \subseteq X$ be dense, let Y be complete, and let $f\colon D \to Y$ be uniformly continuous for the subspace metric on D. Show that there is exactly one continuous $F\colon X \to Y$ with $F|_D = f$, and that this F is uniformly continuous. (For well-definedness, interleave two approximating

sequences into one.)

5. Suppose d is an ultrametric on X in the sense of definition 1.1.10. Show that $\widehat{d}$ is an ultrametric on $\widehat{X}$, and deduce from proposition 1.1.12 that every triangle in the completion of $(\mathbb{Q}, d_p)$ is isosceles.

6. Exhibit a complete metric space Z and an isometric embedding $j\colon (0,1) \to Z$ for which Z is not isometric to the completion of $(0,1)$. Conclude that the density requirement in theorem 2.4.2 cannot be dropped without losing theorem 2.4.3.

3

Draft Material for Chapters 2 to 7

3.1 Compactness

In metric spaces, we can define the concept of sequential compactness:

Definition 3.1.1: Sequentially Compact

A subset $A \subseteq X$ of a metric space (X, d) is *sequentially compact* if for every sequence in A, there exists a convergent subsequence that converges in A

3.1.1 Compactness in compact metric spaces

Theorem 3.1.2: Generalized Heine-Borel Theorem

A subset of a complete metric space is compact if and only if it is closed and totally bounded.

***Proof*:**
p.104 Pugh

Sometimes, the following corrollary is known as generalized Heine Borel. In $\mathbb{R}$:

$$\text{Compact} \Rightarrow \text{closed and bounded}$$

and for the converse in metric spaces:

$$\text{Compact} \Leftrightarrow \text{complete and totally bounded}$$

which is the result of his next corrollary

Corollary 3.1.3: Compactness Of Metric Space (Generlized Heine-Borel)

A metric space is compact if and only if it is complete and totally bounded

***Proof*:**
p.104

summary

[covering] compactness : every open cover has finite subcover

sequentially compcat: every sequence has a convergent subsequence

In $\mathbb{R}$, Heine Borel provides:

$$\text{compact subset} \Leftrightarrow \text{ closed and bounded subset}$$

In metric space:

$$\text{compact subset } \Rightarrow \text{ closed and bounded subset} \qquad \text{compact subset} \not\Leftarrow \text{ closed and bounded subset}$$

Or, if we think about it in terms of the entire space:

$$\text{compact space } \Rightarrow \text{ bounded space} \qquad \text{compact space} \not\Leftarrow \text{bounded space}$$

counter example being $\mathbb{Q} \subseteq \mathbb{R}$

Example 3.1.4: Example

Take $[0, \frac{1}{\pi} - \frac{1}{n}] \cup [\frac{1}{\pi} + \frac{1}{n}, 1]$ for $n \geq 4$. This covers all of $\mathbb{Q} \cap [0, 1]$, but has no fintie subcover. Note that these sets are open in the subspace topology, but not in $\mathbb{R}$.

However:

$$\text{compact space} \Leftrightarrow \text{ complete and totally bounded}$$

and os in complete metric space:

$$\text{compact space} \Leftrightarrow \text{ totally bounded}$$

and using theorem ref:HERE, we get:

$$\text{compact subset} \Leftrightarrow \text{ closed and totally bounded subset}$$

Note that $C^0([a, b], \mathbb{R})$ is complete (a proof in prop ref:HERE). However, the infinite-dimensionality of it means we need another condition for compactness: Using Arzela Ascoli, we get:

$$\text{compact subset} \Leftrightarrow \text{ closed, uniformally bounded, and equicontinuous}$$

For "closed and bounded $\Leftrightarrow$ compact" in a metric space you need:

$$\text{complete, } \sigma\text{-compact, locally compact}$$

which we'll never use in these notes, but it's a fun trivial.

3.2 The Contraction Principle

Completeness has a powerful fixed-point consequence: on a complete metric space, any map that uniformly shrinks distances has exactly one fixed point, and it is reached by iterating the map from any starting point. This is the engine behind many existence-and-uniqueness theorems in analysis, from the inverse function theorem to the existence and uniqueness of solutions of ordinary differential equations.

Definition 3.2.1: Fixed Point

A *fixed point* of a map $f : X \to X$ is a point $p \in X$ with $f(p) = p$.

Definition 3.2.2: Contraction

Let (X, d) be a metric space. A map $f : X \to X$ is a *contraction* if there is a constant k with $0 \le k < 1$ such that

$$d(f(x), f(y)) \le k\, d(x, y) \qquad \text{for all } x, y \in X.$$

Every contraction is Lipschitz, and hence uniformly continuous: given $\epsilon > 0$, the choice $\delta = \epsilon$ makes $d(x, y) < \delta$ force $d(f(x), f(y)) \le k\, d(x, y) < \epsilon$.

Theorem 3.2.3: Banach Contraction Principle

Let (X, d) be a nonempty *complete* metric space and let $f : X \to X$ be a contraction with constant k. Then f has a unique fixed point $p \in X$, and for every $x_0 \in X$ the iterates defined by $x_{n+1} = f(x_n)$ converge to p, with

$$d(x_n, p) \le \frac{k^n}{1-k}\, d(x_0, x_1).$$

***Proof*:**
Fix $x_0 \in X$ and set $x_{n+1} = f(x_n)$. Applying the contraction bound repeatedly,

$$d(x_{n+1}, x_n) = d(f(x_n), f(x_{n-1})) \le k\, d(x_n, x_{n-1}) \le \cdots \le k^n\, d(x_1, x_0).$$

Hence for $m > n$, the triangle inequality and the geometric series give

$$d(x_m, x_n) \le \sum_{i=n}^{m-1} d(x_{i+1}, x_i) \le \left(\sum_{i=n}^{m-1} k^i \right) d(x_1, x_0) \le \frac{k^n}{1-k}\, d(x_1, x_0).$$

Since $0 \le k < 1$, the right-hand side tends to 0 as $n \to \infty$, so (x_n) is a Cauchy sequence. As X is complete, $x_n \to p$ for some $p \in X$.

This limit is a fixed point: because f is continuous, $f(p) = f\big(\lim_n x_n\big) = \lim_n f(x_n) = \lim_n x_{n+1} = p$. Letting $m \to \infty$ in the bound above (the distance is continuous in each argument) yields the stated error estimate.

The fixed point is unique: if $f(p) = p$ and $f(q) = q$, then $d(p, q) = d(f(p), f(q)) \le k\, d(p, q)$, so $(1 - k)\, d(p, q) \le 0$. Since $1 - k > 0$, this forces $d(p, q) = 0$, that is, $p = q$.

3.3 Basic Function Spaces

Example 3.3.1: pointwise convergence

1. here
2. Note that it's possible that a poinwise converging sequence of functions "miss" points. Take

$$f_n(x) : (a, b) \to \mathbb{R} \qquad f_n(x) = x^n$$

then $f_n \to 0$, the zero-function, since every point is indeed eventually convering to 0. However, it seems like there will be a point which is sort of "up there". We need a strong notion to fix that

Definition 3.3.2: Uniform Convergence

Let M and N be arbitrary metric spaces. A sequence of function $f_n : M \to N$ *converges uniformally* to the limit $f : M \to N$ if for each $\epsilon > 0$ there is an N such that for all $n \geq N$, and all $x \in M$,

$$|f_n(x) - f(x)| < \epsilon$$

The function f is the *uniform limit* of the sequence (f_n) and we write

$$f_n \rightrightarrows f \qquad \text{or} \qquad \text{unif} \lim_{n\to\infty} f_n = f$$

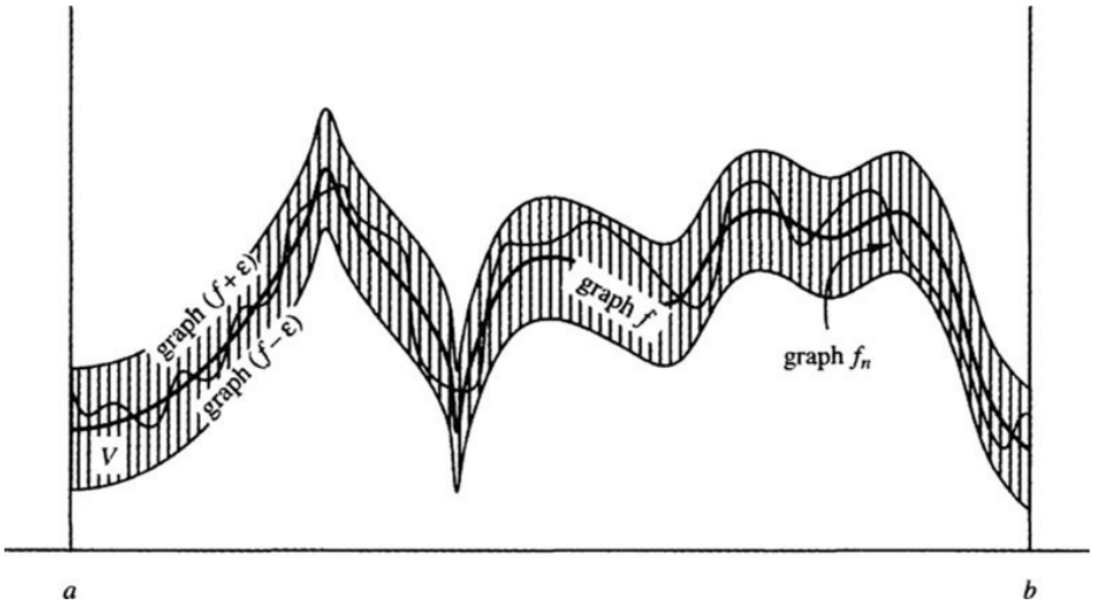

Figure 3.1: uniform convergence visually

Theorem 3.3.3: Uniform Convergence Preserve's Continuity

if $f_n \rightrightarrows f$ and eachf_n is continuous at x_0 then f is continuous at x_0

***Proof*:**
Key is to use the triangle inequality. Details are omitted to be field. Once you choose any $\epsilon > 0$, you know for some $n \geq N$,

$$|f_n(x) - f(x)| < \frac{\epsilon}{3}$$

Then you know for f_N at x_0 there is a $\delta > 0$ st

$$|f_N(x) - f_N(x_0)| < \frac{\epsilon}{3}$$

and then you do the triangle inequality twice, to get split the first equation into 3 equations with f_N being part of it at least once, then use they're all less than $\frac{\epsilon}{3}$, and done

So, with $f_n(x) = x^n$, these function do *not* uniformally converge to 0, but to

$$f(x) = \begin{cases} 0 & \text{if } 0 \leq x < 1 \\ 1 & \text{if } x = 1 \end{cases}$$

Note that the converse is not true: if at every point x_0, f_n is continuous and f is continuous at x_0, this does not imply (f_n) is uniformally convergent:

Example 3.3.4: growing steeple
This is called the growing steeple:

$$f_n(x) = \begin{cases} n^2 x & \text{if } 0 \leq x \leq \frac{1}{n} \\ 2n - n^2 x & \text{if } \frac{1}{n} \leq x \leq \frac{2}{n} \\ 0 & \text{if } \frac{2}{n} \leq x \leq 1 \end{cases}$$

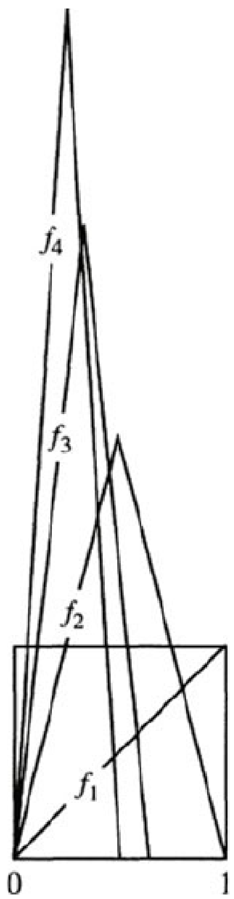

Figure 3.2: growing steeple function

and if you don't like that it goes to infinity, multiply it by $\frac{1}{n}$

Application to function spaces

Definition 3.3.5: Space Of Bounded Functions

Let $C_b([a,b],\mathbb{R})$ represent the set of all bounded functions from $f:[a,b]\to\mathbb{R}$

Then we can define the following norm on this space:

Definition 3.3.6: Sup Norm

Let $f \in C_b$. Then:

$$\|f\| = \sup\{|f(x)||x \in [a,b]\}$$

you should check that this is indeed a norm, and that it will induce a metric as well, and hence a metric topology, and hence the idea of convergence.

In this metric space under this metric, the definition is the same as uniform convergence

Theorem 3.3.7: sup norm convergence if and only if uniformally convergence

Convergence with respect to the sup metric d is equivalent to uniform convergence

***Proof*:**
The two definition's are in fact identical.

Since $d(f_n, f) \to 0$, then for every $x \in [a,b]$ the distnce is becomming smaller, so $|f(x) - f_n(x)|\epsilon$, and so $f_n \rightrightarrows f$.

Theorem 3.3.8: C_b Is A Complete Metric Space

C_b is a complete metric space

***Proof*:**
p.216. If you take any cauchy sequence of functions, then at every point, it's a cuachy sequence in $\mathbb{R}$. use the face that $\mathbb{R}$ is complete while using the sup-norm to complete the proof.

With this, we can show that the space of continuous bounded functions is complete too.

Definition 3.3.9: Continuous Bounded Funcitons

Let $C^0([a,b],\mathbb{R})$ denote the set of continuous functions from $f : [a,b] \to \mathbb{R}$. Note that since the domain is compact, the image is compact, and in $\mathbb{R}$ that implies it's bounded, so $C^0 \subset C_b$.

Corollary 3.3.10: C^0 Is A Compelte Metric Space

C^0 Is A Compelte Metric Space

***Proof*:**
By theorem 3.3.3, every convergent sequence in C^0 lies in C^0, thus C^0 contains it's limits, and thus is closed. so it's a closed subset of a compelte metric space. Thus, by HERE, it's complete

3.4 Application to other concepts

We'll now explore how uniform convergence works with derivatives, integrals, and series

Let

$$F_n = \sum_{i=1}^{n} f_i(x)$$

Then we can say that (F_n) conveges uniformally just like with (f_n) since it's also a sequence of functions.

If we have

$$F_n = \sum_{i=1}^{\infty} f_i(x)$$

Then we can say it converges uniformally if the sequence of partial sums converges uniformally.

Also, we can say the series of absolutes values $\sum |f_k(x)|$ converges then the series $\sum f_k$ *converges absolutely*

Theorem 3.4.1: Weierstrass M-Test

If $\sum M_k$ is a convergetn sereis of constants and if $f_k \in C_b$ satisfies $\|f_k\| \leq M_k$ for all k then $\sum f_k$ converges uniformly and absolutely

***Proof*:**
For n large enough, we can use the triangle enquality on the partial sums to expand out our series:

$$\begin{aligned} d(F_n, F_m) &\leq d(F_n, F_{n-1}) + \cdots + d(F_{m+1}, F_m) \\ &= \sum k = m + 1^n \|f_k\| \leq \sum k = m + 1^n M_k \end{aligned}$$

and thus, since $\sum M_k$ converges, then we can always use it's epsilon for large enough m, n. This shows that (F_n) is cauchy, and so since C_b is compelte, it converges uniformly.

Theorem 3.4.2: Uniform Limits Through Integrals

The uniform limit of Riemann integrable functiosn is Riemann integrable, and hte limit of hte integrals is the integral of hte limit

$$\lim_{n \to \infty} \int_a^b f_n(x)dx = \int_a^b \text{uniflim}_{n \to \infty} f_n(x)dx$$

***Proof*:**
p.218

Note that since all bounded contiuous function are integrable, but not all integrable bounded functions are continuous, this means that the subset of C_b of functions for which this theorem applies is between

C_b and C_0 (since not all bounded functions are integrable. If $\mathcal{R}$ represent all integrable funcitons:

$$C_b \supset \mathcal{R} \supset C^0$$

Now, using the fact that a limit can pass through integrable functions, we can show that infnite sums of of integrated functions is the sme as the integral of the infinite sum of functions:

Theorem 3.4.3: Term By Term Integration Theorem

A uniformly convergetn series of integrable functions $\sum f-k$ can be integrated term-by-term in the sense that

$$\int_a^b \sum_{k=0}^{\infty} f_k(x)dx = \sum_{k=0}^{\infty} \int_a^b f_k(x)dx$$

Proof:
Use the definition, then apply theorem 6.

Now, we move on to differentiable functions

Theorem 3.4.4: Uniform Limit Of Differentiable Functions Is A Differentiable Function

The uniform limit of a sequecne of idfferentiable functiosn is a differentiable function provided that the sequence of derivaties also converges uniformally

Proof:
p.219-220

Assuming f' is continuous makes the proof easy enough using the fundamental theorem of calculus by splitting up $f_n(x) = f_n(a) + \int_a^x f_n'(t)dt$.

For f' not continuous, more to check. Check the book, it's not too bad to follow

This also looks cool: here

Note that this can fail if the derivative doesn't converge uniformally

Example 3.4.5: derivative not converge uniformally
Take

$$f_n(x) = \sqrt{x^2 + \frac{1}{n}}$$

Then this series converges to $f(x) = |x|$, which is not differentiable! The derivatiev converges point-wise, but not uniformally.

Note that in complex analysis, this will not be the case! the uniform limit of a complex differentiable function is complex differentiable, and the sequence of derivatives will automatically be unifomrally convergent to a limit!

LIke before, infinite sums work well

Corollary 3.4.6: Series Of Infinite Sum And Uniform Convergence

A uniformly convergent sereis of differentaible functiosn can be differentiated term-by-term, provided that the derivative series converges uniformly,

$$\left(\sum_{k=0}^{\infty} f_k(x)\right)' = \sum_{k=0}^{\infty} f_k'(x)$$

***Proof*:**
Using the definition of convergence for infinite series, apply theroem 3.4.4 on the partial sums.

3.5 Compactness and equicontinuity

We now want to apply the Heine-Borel theorem to C^0, because checking something is closed and bounded is relatively easy, and for some reason, we want to able to easily find convergent-subsequence of convergent sequences (will come back here once I figure out why TBD)

In C^0, closed and bounded sets are usually not compact. Take

$$B = \left\{f \in C^0([0,1], \mathbb{R}) | \, \|f\| \leq 1\right\}$$

that is, teh set of all functions who's supremum over $[0, 1]$ is 1. This set is not closed: it contains convergent sequences that do not lie inside B, however the value it converges to is *not* continuous

$$f(x) = \begin{cases} 0 & x < 1 \\ 1 & x = 1 \end{cases}$$

This comes down to the fact that $f(x)$ is infinite dimensional: Heine-Borel will applies to finite-dimensional spaces!

To show that C^0 is an infinite dimensional vector-space over $\mathbb{R}$, I'll show that it has an infinite dimensional linearly independent set (I'll not be showing it's a basis here, but since a set exists, the basis must at least have this cardinatliy) Take the set of "smooth bump" funtions, where f_0 start it's bump at 0 and ends at 1. Then $\{f_n | n \in \mathbb{Z}\}$ is a linearly independent set of continuous (even smooth) functions. Thus, the space must at least have a countably infinite dimension.

Now back to finding the appropriate conditions for the Heine-Borel property. To do this, we need a new condition on our functions:

Definition 3.5.1: [Uniform] Equicontinuous

A sequence of functions (f_n) in C^0 is **equicontinuous** if

$$\forall \epsilon > 0, \exists \delta > 0 \text{ such that } |s - t| < \delta, \forall n \in \mathbb{N} \Rightarrow |f_n(s) - f_n(t)| < \epsilon$$

The functions f_n are *equally continuous.*

Note that the δ depends on teh ϵ, but *not* on the n I put [uniform] in the definition because there are definition of equicontinuity which are not "uniform", that is, you pick a point for which the conditions applies:

Definition 3.5.2: Point-Wise Equicontinuous

A sequence (f_n) in C^0 is *pointwise equicontintuous* if

$$\forall \epsilon > 0 \forall x \in [a,b], \exists \delta > 0 \text{ such that } |x-t| < \delta, n \in \mathbb{N}, |f_n(x) - f_n(t)| < \epsilon$$

From here on out, we will say equicontinuity instead of uniform equicontinuity.

Example 3.5.3: Example

1. The set of function $f : \mathbb{R} \to \mathbb{R}$, $f_n(x) = \frac{x}{n}$ are equicontinuous. However, they are not uniformally continuous.
2. Take $f : [0,1] \to \mathbb{R}$, $f_n(x) = n$. That is, a constant function. Then $\{f_n\}$ is an equicontinuous sequence. However $\{f_n\}$ is not a convergent sequence (and so not uniformally convergent).
3. let $\{f\}$ be a equicontinuous set with a single function. Then f is uniformally continuous by definition. In this sense, equicontinuity is intimately related to uniform continuity.

<++>

This definition can also be used to define an equicontinuous set:

Definition 3.5.4: Equicontinuous Set

The set $\mathcal{E} \subset C^0$ is equicontinuous if

$$\forall \epsilon > 0, \exists \delta > 0 \text{ such that } |s-t| < \delta \text{ and } f \in \mathcal{E} \Rightarrow |f_(s) - f(t)| < \epsilon$$

Essentially, if a function is continuous, than for all $\epsilon > 0$, there exists an open neigbhorhood U such that $d(f(x), f(y)) < \epsilon$ for all $y \in U$. Equicontinuity means that a single neighborhood U can be chosen that will work for all the functions f in the collection $\mathcal{E}$.

You might recall Bolzano-Weierstrass theorem:

Theorem 3.5.5: Bolzano-Weierstrass Theorem

Let $M = \mathbb{R}^n$. Then every bounded sequence has a convergent subsequence.

***Proof*:**
Here is the sketch of the proof.

Since it's bounded, put it in a ball, and take the closure of the ball. Then it's closed. So it's compact. Take the cover: $B_1 = \{B_1(x) | x \in \mathbb{R}^n\}$. Then it has a finite sub-cover B_1'. Then one of the ball must have infinitely many points. Let's say $b_1' \in B_1'$. Then cover this ball with $B_{1/2}(x)$ balls, and repeat the same process. Keep doing this for every n. Then you can pick a sub-sequence from each $B_{1/n}(x)$ which will converge.

We'll be proving the analogous of this with equicontinuous sequences in C^0:

Theorem 3.5.6: Arzela-Ascoli Theorem

Let M be a compact metric space. Every uniformally bounded equicontinuous sequence of functions in $C^0(M, \mathbb{R})$ has a uniformly convergent subsequence

***Proof*:**
p.226. Fun to follow!

Notice how this relates to compactness: if every subsequence of f_n converges, then the closure of $\{f_n | n \in \mathbb{N}\}$ is compact!

The last part of the proof also shows that you can weaken the condition a bit, and say that if you converge on a dense subset, then you're uniformally convergent.

Corollary 3.5.7: Arzela-Ascoli Propagation Theorem

Pointwise convergence of an equicontinuous sequence of functions on a dense subset of the domain *propagates* to uniform convergence on the whole domain

***Proof*:**
last part of the proof for Ascoli

Another thing that I don't konw where to put, but I think is neat to consider:

1. Let's say that you have a sequence of function where $f_n \to f$ pointwise. you also have sequences of functions $f_{n,m} \to f_n$, that is, every f_n has a sequnce of function's converging to it. Then is there a subseuqence of $f_{n,m}$ (for varaible n, m) that converges to f?

<++>

If you're derivative is bounded, we cna actually prove this with a much weaker claim: only 1 point must be convergent for the whole thing to be uniformally continuous!!

Corollary 3.5.8: Differentiable Convergent Subsequence

Assume that $f_n : [a, b] \to \mathbb{R}$ is a sequence of differentiable functions whose derivatives are uniformly bounded. If for one point x_0, the sequence $(f_n(x_0))$ is bounded as $n \to \infty$, then the sequence (f_n) has a subsequence that converges uniformly on the whole interval $[a, b]$.

***Proof*:**
Let M be the bound of $|f_n'(x)|$ for all $n \in \mathbb{N}$ and $x \in [a, b]$. Equicontinuity of (f_n) then follows from the MVT:

$$|s - t| < \delta \Rightarrow |f_n(s) - f_n(t) = |f_n'(\alpha)||s - t| \leq M\delta$$

and then set $\delta = \frac{\epsilon}{(M+1)}$. THis shows that (f_n) is equicontinuous!

Now to show that the sequence is uniformally bounded, let C be a bound for $|f_n(x_0)|$ for all $n \in \mathbb{N}$. THen

$$\begin{aligned} |f_n(x) \leq |f_n(x) - f_n(x_0)| + f_n(x_0)| &\leq M|x - x_0| + C \\ &\leq M|b - a| + C \end{aligned}$$

showing that the sequence (f_n) is bouded in C^0! Since it's uniformally bounded and equicontinuous, by Arzela-Ascoli's theorem (3.5.6), $f(f_n)$ has a uniformally convergent subsequence!

As a quick aside, this theorem has two uses in other domain we will not discuss;

1. A sequecne of solutions to a continuous ordinary differential equation in $\mathbb{R}^M$ has a subsequence that converges to a limit, and that limit is also a solutionof the ODE
2. A sequence of complex analytic functions that converges pointwise, converges uniformly (on compact subsets of the domain of definition) and the limit is complex analytic.

However, we'll be using Arzela-Ascoli for Heine-Borel:

Theorem 3.5.9: Heine-Borel Theroem In A Function Space

A subset $\mathcal{E} \subseteq C^0$ is compact if and only if it is closed, bounded, and equicontinuous

***Proof*:**

($\Rightarrow$) Since C^0 is a complete metric space, it is totally bounded (it is automatically closed since C^0 is clopen). This means for some $\epsilon > 0$, ew can cover $\mathcal{E}$ with finitely many balls of size $\epsilon/3$

$$\mathcal{C} = \left\{B_{\epsilon/3}(f_k) | \text{appropriate } f_k\right\}$$

Note that each f_k is uniformally continuous since the domain is compact. Thus, there exists a $\delta > 0$ st

$$|s - t| < \delta \Rightarrow |f_k(s) f_k(t)| < \frac{\epsilon}{3}$$

Now, if $f \in \mathcal{E}$, the for some k we hvae $f \in B_{\epsilon/3}(f_k)$, and $|s - t| < \delta$ will give us

$$|f(s) - f(t)| \leq |f(s) - f_k(s)| + (f_k(s) - f_k(t)| + |f_k(t) - f(t)| < 3\frac{\epsilon}{3} = \epsilon$$

showing $\mathcal{E}$ is equicontinuous.

($\Leftarrow$) Assume $\mathcal{E}$ is closed, bounded, and equicontinuous. Then if (f_n) is a sequence in $\mathcal{E}$, by Arzela-Ascoli's theorem, some subsequence converges uniformly to a limit. This limit lies in $\mathcal{E}$ since $\mathcal{E}$ is clsoed. Thus, by ref:HERE, $\mathcal{E}$ is compact

3.6 Uniform Approximation in C^0

We now move onto a really interesting part of analysis: Given a continuous but nondifferentiable function f (so, any function in C^0), we we can "arbitrarily" approximate f with a smooth function

g! Even better, this smooth function will be a polynomial!!

In order for this to be possible, it would mean that for any $f \in C^0$, there is a polynomial close to it. That is what we're about to prove:

Theorem 3.6.1: Weierstrass Approximation Theorem

The set of polynomials is dense in $C^0([a,b],\mathbb{R})$

Note that since the function is continuous on a compact set, it is uniformally continuous (a fact we'll use in the 2nd proof).

Decomposing what density means for a moment: for each $f \in C^0$, and each $\epsilon > 0$, there is a polynomial function $p(x)$ such that for all $x \in [a,b]$

$$|f(x) - p(x)| < \epsilon$$

meaning that we can arbitrarily approximate $f(x)$!!

***Proof*:**
#1 The key idea to building $p(x)$ from f is to sample values of f and recombining them in a clever way.

Without loss of generality, let's say $[a,b] = [0,1]$. For each $n \in \mathbb{N}$, consider

$$p_n(x) = \sum_{k=0}^{n} \binom{n}{k} c_k x^k (1-x)^{n-k}$$

where $c = f\left(\frac{k}{n}\right)$ and $\binom{n}{k}$ is the binomial coefficient. This polynomial is called the **Bernstein polynomial**. We'll show that (p_n) converges uniformally to f as $n \to \infty$, i.e., $(p_n) \rightrightarrows f$.

***Proof*:**
#2 This proof uses convolution. to link the nice properties of polynomials with the less nice propertries of general continuous functions.

Without loss of generality, let's let $[a,b] = [0,1]$ (they are homeomorphic). Let $f \in C^0([0,1],\mathbb{R})$. What we'll do is manipulate this functions to a nicer homeomorphic equivalent, and approximate that function with a polynomial.

Let $g(x) = f(x) - (mx + b)$, where

$$m = \frac{f(1) - f(0)}{1} = f(1) - f(0) \qquad \text{and} \qquad b = f(0)$$

Since we're adding a polynomial to $f(x)$, $g \in C^0$. Notice that

$$g(0) = f(0) - (f(1) - f(0))\,0 - f(0) = 0 \qquad g(1) = f(1) - f(1) + f(0) - f(0) = 0$$

so $g(0) = 0 = g(1)$. This shows that given any f, it is equivalent to consider f in the "form" of g (the two being homeomorphic), so let $f(0) = 0 = f(1)$. In our construction, we'll sometimes need the domain to extend beyond $[0,1]$ for technical reasons, so let's say $f(x) = 0$ for $x \in \mathbb{R} - [0,1]$ (this is a smooth extension with zero measure, so nothing bad can come from this).

Next, we will construct the nice function we'll convolute with. Let

$$\beta_n(t) = b_n(1-t^2)^n \qquad -1 \le t \le 1$$

where the constant b_n is chosen so that $\int_{-1}^{1} b_n(t)dt = 1$. For $0 \le x \le 1$, set

$$P_n(x) = \int_{-1}^{1} f(x+t)\beta_n(t)dt$$

This is a weighted average of the value of f using the weight function β_n. This is where we use convlution to get the nice properties of $\beta_n(t)$ to f. We claim that P_n is a polynomial, and $P_n(x) \rightrightarrows f(x)$ as $n \to \infty$.

To check that P_n is a polynomial, we use a change of variables: $u = x+t$. Then for $0 \le u \le 1$ we have

$$P_n(x) = \int_{x-1}^{x+1} f(u)\beta_n(u-x))du = \int_{0}^{1} f(u)\beta_n(u-x)du$$

where the last integral is because $f = 0$ outside $[0,1]$. The function $\beta_n(u-x) = b_n(1-(u-x^2))^n$ is a polynomial in x whose coefficients are polynomials in u. We can pull out all the x terms, since the integral is with respect to x, and since the integral is definite with concrete endpoints, it is a constant, and so we have a polynomial!! In other words, we managed to "eliminate" the complexity of f.

To check that $P_n \rightrightarrows f$ as $n \to \infty$, we need to estimate $\beta_n(t)$. We claim that if $\delta > 0$ then

$$\beta_n(t) \rightrightarrows 0 \text{ as } n \to \infty \text{ and } \delta \le |t| \le 1$$

Notice that $\delta > 0$, so $0 \le |t| \le 1$, and so we are ignoring the fact that $x = 0$ diverges.

You can prove this directly, but I'll skip it for now (p.233)

Now, we'll deduce $P_n \rightrightarrows f$. Let $\epsilon > 0$. Since f is uniformally continuous, let $\delta > 0$ such that $|t| < \delta$ imples $f(x+t) - f(x)| < \frac{\epsilon}{2}$. Since β_n has integral 1 on $[-1,1]$, we can slip in β_n in the following equality:

$$\begin{aligned}
|P_n(x) - f(x)| &= \left|\int_{-1}^{1} \left(f(x+t) - f(x)\right)\beta_n(t)dt\right| \\
&\le \int_{-1}^{1} |f(x+t) - f(x)|\beta_n(t)dt \\
&= \int_{|t|<\delta} |f(x+t) - f(x)|\beta_n(t)dt + \int_{|t|\ge\delta} |f(x+t) - f(x)|\beta_n(t)dt \\
&< \epsilon/2 + 2M\int_{|t|\ge\delta} B_n(t)dt
\end{aligned}$$

where the 1st integral is less than $\frac{\epsilon}{2}$ becuase HERE, and the 2nd because HERE. Since δ is positive, and $\beta_n \rightrightarrows 0$, then the 2nd integral becomes $\frac{\epsilon}{2}$ as well.

Thus, $P_n \rightrightarrows f$ – as we claimed

With this, we can prove the general Stone-Weierstrass theroem on compact metric spaces: that is, change $[0,1]$ into a general comapct metric space.

To do this, we'll first define an algebra on the functions. Let $A \subseteq C^0(M, \mathbb{R})$ such that if $f, g \in A$, $c \in \mathbb{R}$, then

$$f + g \in C^0(M, \mathbb{R}),\ cf \in C^0(M, \mathbb{R}),\ f \cdot g \in C^0(M, \mathbb{R})$$

where each operation is defined pointwise (ex. $(f + g)(x) = f(x) + g(x)$). For examples, polynoimals form an algebra.

The function algbera *vanishes at point* p if $f(p) = 0$ for all $f \in A$. FOr example, A with polynoimals with zero constatn term vanishes at $x = 0$.

THe function algebra *seperates poitns* if for each pari of distinct points $p_1, p_2 \in M$, ther eis a function $f \in A$ such that $f(p_1) \neq f(p_2)$. FOr example, the function algebra of all tirgonometric polynoimals seperates the points $[0, 2\pi)$ and vanishes nowhere

Theorem 3.6.2: Stone-Weierstrass Theorem

If M is a compact metric space and A is a function-algebra in $C^0(M, \mathbb{R})$ that vanishes nowhere and separates points, then A is dense in $C^0(M, \mathbb{R})$

Note that the Weierstrass theorem is a special case of this one, but it will be used to prove this one

***Proof*:**
We first prove 2 lemmas to help us

If A vanisehs nowhere and seperates points then there exists $f \in A$ with specified values at any pari of disitnct ponts

Proof. □

1. power series: definition, integration, differentiation
2. define analytic
3. Weierstrass aproximation theroem(set of polynomials is dense in $C^0([a, b], \mathbb{R})$)
4. Stone-Weierstrass Theorem
5. Banach Contraction Principle?
6. Picard's Theorem?

<++>

3.7 Nowhere Differentiable Continuous Functions

Theorem 3.7.1: Nowhere Differentiable Continuous Functions

There exists a continuous function $f : \mathbb{R} \to \mathbb{R}$ that has a derivative at no point whatsoever

***Proof*:**
We'll construct a function by taking an infinite series of sawtooth functions wich will sum to this:

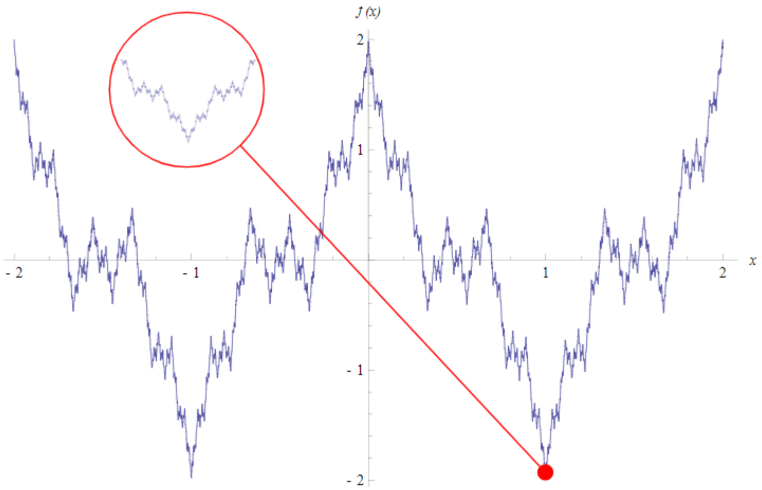

Figure 3.3: Nowehere Differential, yet Continuous Functions

Let the letters k, m, n dentoes integers. Start with the sawtooth function $\sigma_0 : \mathbb{R} \to \mathbb{R}$ defined as

$$\sigma_0(x) = \begin{cases} x - 2n & \text{if } 2n \leq x \leq 2n+1 \\ 2n+2-x & \text{if } 2n+1 \leq x \leq 2n+2 \end{cases}$$

Note that σ_0 is periodic with period 2

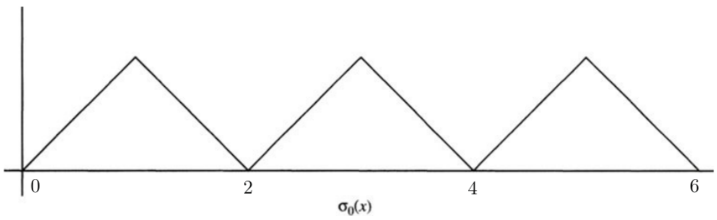

Figure 3.4: σ_0

Next, defined the next sawtooth functions by compression σ_0:

$$\sigma_k(x) = \left(\frac{3}{4}\right)^k \sigma_0(4^k x)$$

the period will be $\frac{2}{4^k}$, since $\sigma(x) = \sigma(x + m\left(\frac{2}{4^k}\right))$ for any $m \in \mathbb{Z}$.

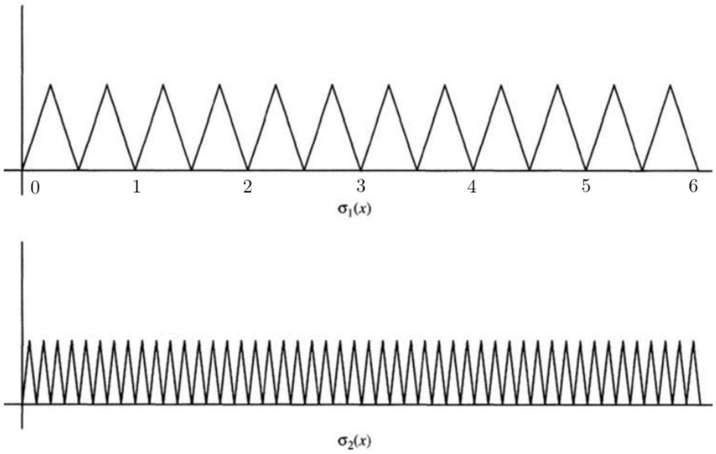

Figure 3.5: two examples of the itteration

By the M-test, $\sum \sigma_k(x0)$ converges uniformly to a limit f since $|\sigma_k| \leq \left(\frac{3}{4}\right)^k$ and

$$f(x) = \sum_{k=0}^{\infty} \sigma_k(x)$$

is continuous. We'll show that f is nowhere differentiable, that is

$$\lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

does not exist. Since this must work no matter how $h \to 0$, pick $\delta_m = \frac{1}{2 \cdot 4^m}$. Notice that $\delta_m \to 0$ as $m \to \infty$. We'll show that this limit does not exist:

$$\lim_{m \to 0} \frac{f(x \pm \delta_m - f(x)}{\delta_m}$$

where the $\pm$ is because we want this limit to reprent approach from either the left or right.

Using proposition 3.4.6, we get the derivative is

$$\sum_{k=0}^{\infty} \frac{\sigma_k(x \pm \delta_m) - \sigma(x)}{\delta_m}$$

We'll split this sum into 3 cases: $k > m$, $k = m$, $k < m$

($k > m$) Since $\delta_m = \frac{2}{4^k}$

$$\delta_m = \frac{1}{2}\frac{1}{4^m} = m\sigma_k \qquad m \in \mathbb{Z}$$

and thus

$$\sigma_k(x \pm \delta_m) - \sigma_k(x) = \sigma_k(x) - \sigma_k(x) = 0$$

meaning these terms do not contribute to the sum. We can essentially re-write our sum to

$$\sum_{k=0}^{n-1} \frac{\sigma_k(x \pm \delta_m) - \sigma(x)}{\delta_m}$$

($k = m$) σ_m will be monotone on either $[x, x+\delta_m]$ or $x - \delta - m, x]$:

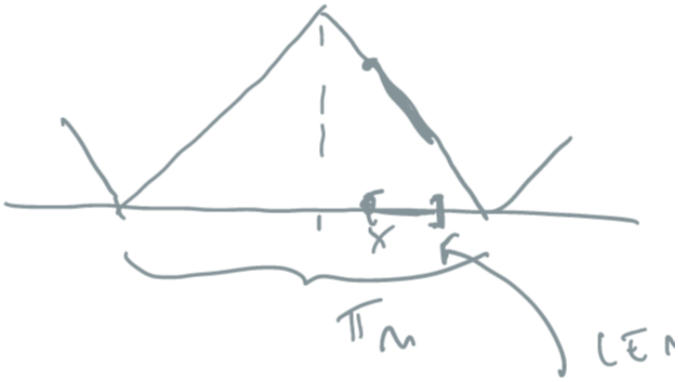

Figure 3.6: Demonstration of monotonicity

and therefore

$$|frac\sigma_m(x+\delta_m) - \sigma_m(x)\delta_m| = 3^m$$

($k < m$) using what we've just learn from the previous exercice, we can crudely estimate each σ_k:

$$\left|\frac{\sigma_k(x \pm \delta_m) - \sigma_k}{\delta_m}\right| \leq 3^k$$

giving us

$$\sum k < m \left|\frac{\sigma_k(x \pm \delta_m) - \sigma_k}{\delta_m}\right| \leq \sum_{k<m} 3^k = \frac{3^m - 1}{3 - 1}$$

Thus, this gives us

$$\lim_{h \to 0} \frac{f(x+h) - f(x)}{h} = \lim_{m \to \infty} \sum_{k=0}^{\infty} \frac{\sigma_k(x \pm \delta_m) - \sigma(x)}{\delta_m} = \lim_{m \to \infty} 0 \pm 3^m + (3) \geq 3^m - \frac{3^m - 1}{2} = \frac{3^m}{2} + \frac{1}{2}$$

which diverges as $m \to \infty$! Therefore, this function is *not* differentiable at any point

3.8 Baire Theory

This construction might look like a particular case we need to construct to find, but it turns out that almost all continuous functions are nowhere differentiable! If you define your notion of probability right, then the chance of picking a continuous differentiable function form the set of continuous function is 0%! The way we'll make this rigorous is by taking a detour into Baire spaces.

A set D is dense in M if $\overline{D} = M$. Equivalently, it is dense in M if D for any open $W \subseteq M$, $W \cap M \neq \emptyset$. The intersection of dense sense need not be dense (ex. $\mathbb{Q}$ and $\mathbb{Q}^c$ in $\mathbb{R}$). However, if

U, V are *open*-dense sets in M, then $U \cap V$ will also be open-dense in M. If $W \subseteq M$ is an arbitrary open subset. Then $U \cap W$ is also open, and so by denseness of V, $V \cap (U \cap W)$ is also non-empty! re-arranging using

3.9 Cardinality quick

I'm not sure where to add this yet, but I thought it is a good exercise to appreciate the oddity of $\aleph_1$

Pove that C^0 and $\mathbb{R}$ have the same cardinality

It is clear that C^0 has at least as many function as real numbers since every constant function is continuous, so it comes down to show there are no more. This is exercise 6 in Pugh's book, and its marked with a double star.

Found this too

https://math.stackexchange.com/questions/477/cardinality-of-set-of-real-continuous-functions

4

What Next?

The theorems of this book are tools, and each one is sharpest in a subject that lies just past its statement.

Dropping the metric. Completeness and the contraction principle need a distance, but compactness, connectedness and continuity do not: they can be stated using open sets alone, and in that generality they apply to spaces no metric describes. *Topology* [5] makes that move and is the point-set foundation of most of the series. Munkres' *Topology* is the standard companion.

Adding linear structure. The space $C(K)$ studied here is a normed vector space that happens to be complete, and that combination has a name: a Banach space. Taking it as the primitive object gives the Hahn-Banach theorem, the open mapping and closed graph theorems (both proved from Baire category), and the spectral theory of operators. That is *Functional Analysis* [6]. Rudin's *Functional Analysis* and Brezis' *Functional Analysis, Sobolev Spaces and Partial Differential Equations* are the two standard routes.

Fixed points as solutions. The contraction principle is the engine behind the existence and uniqueness theorem for differential equations: Picard iteration is a contraction on a space of continuous functions, and its fixed point is the solution. *Ordinary Differential Equations* [4, Picard-Lindelöf Existence and Uniqueness] carries this out. Hartman's *Ordinary Differential Equations* is the thorough reference.

Measuring instead of approximating. $C(K)$ is complete under the sup norm, but not under the integral norms that analysis actually wants, and repairing that requires rebuilding integration from measure theory. *Measure Theory* [3] does so; Folland's *Real Analysis* is the standard text.

Metric spaces as geometry. This book treats a metric as a tool for limits. Treating it instead as geometry, and asking which spaces look like non-positively curved manifolds at large scale, opens metric geometry and geometric group theory. Burago, Burago and Ivanov's *A Course in Metric Geometry* and Bridson and Haefliger's *Metric Spaces of Non-Positive Curvature* are the entry points.

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